

Kinora Upper Extremity AI Clinical Orientation Database

Structured reference for RAG / AI-assisted physiotherapy consultation

Upper Extremity: Anatomy, Examination, Pathology & Rehabilitation

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Disclaimer

This document is an educational orientation resource for Kinora AI clinical consultation support. It is NOT a substitute for professional clinical judgment, direct patient examination, or licensed medical/physiotherapy care. Content reflects established musculoskeletal medicine and sports rehabilitation concepts but must be verified against current peer-reviewed literature, local protocols, and individual patient presentation. Always screen for red flags requiring urgent medical referral. Diagnostic sensitivity/specificity values are approximate and vary by study.

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1. Bones

RECORD: Clavicle

ID: BONE-UE-001

Name: Clavicle

Latin Name: Clavicula

Alternative Names: Collarbone

Region: Shoulder girdle / upper thorax

Body Segment: Upper extremity — proximal girdle

Bone Type: Long bone (modified S-shaped)

Shape: S-shaped; medial two-thirds convex anteriorly, lateral third concave anteriorly

Surface Anatomy: Subcutaneous along entire length; palpable from sternoclavicular to acromioclavicular joint

Anatomical Landmarks: Sternal end, acromial end, conoid tubercle, trapezoid line, subclavian groove

Articulations: Sternoclavicular (SC), acromioclavicular (AC), costoclavicular ligament to 1st rib

Muscle Attachments: Trapezius, sternocleidomastoid, subclavius, pectoralis major (clavicular head), deltoid

Ligament Attachments: AC ligaments, coracoclavicular (conoid/trapezoid), costoclavicular, interclavicular

Tendon Attachments: Indirect via muscular attachments; no major tendons insert on clavicle

Blood Supply: Suprascapular artery, thoracoacromial artery, nutrient artery from supraclavicular branch

Venous Drainage: Companion veins to suprascapular and thoracoacromial systems

Ossification: Intramembranous; two primary centers (medial ~5th fetal week, lateral ~8th week); fuses ~25 years

Biomechanical Function: Strut maintaining scapula lateral to thorax; transmits axial load from UE to trunk; allows scapular rotation

Clinical Importance: Common fracture in falls and contact sports; risk to brachial plexus and subclavian vessels medially

Common Fractures: Midshaft (most common), lateral (AC disruption), medial (rare, high-energy)

Healing Time: 6–12 weeks typical; non-union rare midshaft, more common lateral third

Imaging Recommendations: AP clavicle and cephalic tilt; CT for comminution/shortening; consider angiography if vascular concern

References: Rockwood & Green Fractures in Adults; Neer AC joint classification; see PubMed [clavicle fracture management]

RECORD: Scapula

ID: BONE-UE-002

Name: Scapula

Latin Name: Scapula

Alternative Names: Shoulder blade

Region: Posterior thorax — shoulder girdle

Body Segment: Upper extremity — proximal girdle

Bone Type: Flat bone

Shape: Triangular flat bone with spine, acromion, coracoid process, glenoid fossa

Surface Anatomy: Spine palpable posteriorly; acromion at lateral shoulder tip; coracoid deep to deltopectoral groove

Anatomical Landmarks: Glenoid fossa, acromion, coracoid process, scapular spine, supraspinous/infraspinous fossae, subscapular fossa

Articulations: Glenohumeral (via glenoid), acromioclavicular, scapulothoracic (muscular glide)

Muscle Attachments: Rotator cuff (SS, IS, TM, Subsc), deltoid, trapezius, serratus anterior, rhomboids, levator scapulae, teres major/minor, biceps short head, coracobrachialis, latissimus dorsi

Ligament Attachments: Glenohumeral, AC, coracoacromial, coracoclavicular (to coracoid)

Tendon Attachments: Rotator cuff tendons insert on greater/lesser tubercle region via humerus; indirect scapular muscle origins

Blood Supply: Suprascapular, circumflex scapular, dorsal scapular arteries

Venous Drainage: Suprascapular and axillary venous plexus

Ossification: Multiple ossification centers; acromion fuses ~22 years

Biomechanical Function: Mobile base for glenohumeral joint; scapulothoracic rhythm (~2:1 GH:ST ratio in elevation)

Clinical Importance: Scapular dyskinesis linked to impingement, instability, rotator cuff pathology

Common Fractures: Body, glenoid (Bankart fragment), acromion, coracoid (avulsion in instability)

Healing Time: 6–12 weeks; glenoid fractures may require surgical fixation

Imaging Recommendations: Scapular Y-view, axillary lateral; CT for glenoid; MRI for associated soft tissue

References: Kibler scapular dyskinesis; see PubMed [scapular fracture glenoid]

RECORD: Humerus

ID: BONE-UE-003

Name: Humerus

Latin Name: Humerus

Alternative Names: Upper arm bone

Region: Arm

Body Segment: Upper extremity — arm

Bone Type: Long bone

Shape: Proximal head/neck/tubercles; shaft; distal condyles and epicondyles

Surface Anatomy: Greater tubercle lateral in neutral rotation; medial epicondyle palpable at elbow

Anatomical Landmarks: Head, anatomical/surgical neck, greater/lesser tubercles, bicipital groove, deltoid tuberosity, medial/lateral epicondyles, olecranon/coronoid/radial fossae, capitulum, trochlea

Articulations: Glenohumeral, elbow (trochlea-ulna, capitulum-radius), proximal radioulnar

Muscle Attachments: Deltoid, rotator cuff, pectoralis major, latissimus dorsi, teres major, coracobrachialis, brachialis, triceps, brachioradialis, common flexor/extensor origins

Ligament Attachments: Glenohumeral, coracohumeral; ulnar collateral (to medial epicondyle region)

Tendon Attachments: Rotator cuff, biceps long head (supraglenoid), triceps (olecranon)

Blood Supply: Anterior/posterior circumflex humeral, deep brachial (profunda), nutrient artery

Venous Drainage: Brachial and axillary veins

Ossification: Primary center shaft; proximal/distal epiphyses fuse ~16–20 years

Biomechanical Function: Lever for elbow/shoulder; torsion resistant design; epicondyles as tendon pulleys

Clinical Importance: Fractures risk radial nerve (spiral groove); pediatric supracondylar risks brachial artery

Common Fractures: Proximal (Neer classification), shaft (radial nerve palsy), supracondylar (pediatric), epicondyle avulsion

Healing Time: 8–12 weeks shaft; proximal varies with displacement/neck fracture

Imaging Recommendations: AP/lateral shoulder or humerus; CT for articular; MRI for RCT with proximal fracture

References: Neer proximal humerus; AO classification; see PubMed [humeral shaft radial nerve]

RECORD: Radius

ID: BONE-UE-004

Name: Radius

Latin Name: Radius

Alternative Names: Radial bone

Region: Forearm — lateral

Body Segment: Upper extremity — forearm

Bone Type: Long bone

Shape: Proximal head/neck/tuberosity; curved shaft; distal styloid and articular surface

Surface Anatomy: Radial styloid palpable at anatomical snuffbox region

Anatomical Landmarks: Head, neck, radial tuberosity, Lister tubercle, styloid process, ulnar notch

Articulations: Proximal radioulnar, radiocapitellar, distal radioulnar, radiocarpal

Muscle Attachments: Biceps, supinator, pronator teres/quadratus, flexor pollicis longus, abductor pollicis longus, extensors

Ligament Attachments: Annular ligament; radioulnar ligaments; extrinsic wrist ligaments at distal radius

Tendon Attachments: Biceps insertion at tuberosity; EPL crosses Lister tubercle

Blood Supply: Radial artery (anterolateral), anterior/posterior interosseous branches

Venous Drainage: Radial and interosseous veins

Ossification: Epiphyses fuse ~14–17 years

Biomechanical Function: Rotates around ulna in pronation/supination; load transfer at wrist

Clinical Importance: Distal radius fracture most common UE fracture in adults; Essex-Lopresti if DRUJ disrupted

Common Fractures: Distal radius (Colles/Smith), radial head (Mason), shaft (Galeazzi)

Healing Time: 6–8 weeks distal; radial head depends on displacement

Imaging Recommendations: Wrist/forearm AP/lateral; CT for intra-articular; MRI for TFCC/DRUJ

References: AO distal radius; Mason radial head; see PubMed [distal radius fracture]

RECORD: Ulna

ID: BONE-UE-005

Name: Ulna

Latin Name: Ulna

Alternative Names: Ulnar bone

Region: Forearm — medial

Body Segment: Upper extremity — forearm

Bone Type: Long bone

Shape: Proximal olecranon/coronoid/trochlear notch; triangular shaft; distal styloid (small)

Surface Anatomy: Olecranon posterior elbow; shaft subcutaneous medially

Anatomical Landmarks: Olecranon, coronoid process, trochlear notch, ulnar tuberosity, styloid process

Articulations: Humeroulnar, proximal/distal radioulnar

Muscle Attachments: Triceps, brachialis, flexor/pronator group, extensor carpi ulnaris

Ligament Attachments: UCL elbow; annular; interosseous membrane; TFCC ulnar attachment

Tendon Attachments: Triceps at olecranon; FCU at olecranon/medial epicondyle region

Blood Supply: Ulnar artery, anterior/posterior interosseous

Venous Drainage: Ulnar and interosseous veins

Ossification: Epiphyses fuse ~14–17 years

Biomechanical Function: Stable hinge partner at elbow; DRUJ rotation axis medial

Clinical Importance: Olecranon fracture affects elbow extension; Monteggia = ulnar shaft + radial head dislocation

Common Fractures: Olecranon, coronoid, ulnar shaft (Monteggia/Nightstick)
Healing Time: 8–12 weeks; olecranon often surgical if displaced
Imaging Recommendations: Elbow AP/lateral; forearm views for Monteggia; CT for coronoid
References: Regan-Morrey coronoid; see PubMed [olecranon fracture fixation]

RECORD: Scaphoid

ID: BONE-UE-006
Name: Scaphoid
Latin Name: Os scaphoideum
Alternative Names: Navicular (historical, avoid confusion with foot)
Region: Wrist — proximal carpal row, radial side
Body Segment: Hand/wrist
Bone Type: Short bone (carpal)
Shape: Boat-shaped; tubercle on palmar surface
Surface Anatomy: Anatomic snuffbox floor between EPL and APL/EPB
Anatomical Landmarks: Tubercle, waist (narrowest), proximal pole, distal pole
Articulations: Radius, lunate, capitate, trapezium, trapezoid
Muscle Attachments: None direct; extrinsic tendons cross nearby
Ligament Attachments: Scapholunate interosseous ligament (critical); radioscaphocapitate; scaphotrapezial
Tendon Attachments: None direct
Blood Supply: Radial artery branches (dorsal carpal arch); retrograde flow — proximal pole avascular risk
Venous Drainage: Dorsal/palmar carpal venous plexus
Ossification: Single center; ossifies ~4–6 years; fuses late adolescence
Biomechanical Function: Primary wrist extension/radial deviation stabilizer; load transfer radius to distal carpus
Clinical Importance: Scaphoid fracture = AVN risk (proximal pole); missed fracture leads to SNAC wrist
Common Fractures: Waist (most common), proximal pole (high AVN), tubercle
Healing Time: 8–12 weeks waist; 12–24+ weeks proximal; high non-union rate
Imaging Recommendations: Scaphoid series (4 views); MRI most sensitive early; CT for union assessment
References: Russe classification; Herbert screw fixation; see PubMed [scaphoid fracture MRI]

RECORD: Lunate

ID: BONE-UE-007
Name: Lunate
Latin Name: Os lunatum
Alternative Names: Semilunar bone
Region: Wrist — proximal carpal row, central
Body Segment: Hand/wrist
Bone Type: Short bone (carpal)
Shape: Crescent-shaped; convex proximally for radius
Surface Anatomy: Deep; palpable indirectly via wrist crease central
Anatomical Landmarks: Palmar and dorsal surfaces; no prominent tubercles
Articulations: Radius, scaphoid, triquetrum, capitate, hamate
Muscle Attachments: None direct
Ligament Attachments: SL ligament, LT ligament, radiolunate, ulnolunate (via TFCC)

Tendon Attachments: None direct
Blood Supply: Palmar/dorsal carpal branches; variable — Kienböck disease (AVN)
Venous Drainage: Carpal venous plexus
Ossification: Ossifies ~4 years
Biomechanical Function: Central column of wrist; force transmission; lunate tilt reflects carpal instability
Clinical Importance: Perilunate dislocation; Kienböck AVN; SL tear alters lunate orientation
Common Fractures: Fractures rare; dislocation/perilunate injury more common
Healing Time: Variable; AVN management prolonged
Imaging Recommendations: Wrist MRI for AVN/SL; plain films for alignment (DISI/VISI)
References: Lichtman Kienböck staging; Mayfield perilunate; see PubMed [Kienböck disease]

RECORD: Capitate

ID: BONE-UE-008
Name: Capitate
Latin Name: Os capitatum
Alternative Names: Os magnum
Region: Wrist — distal carpal row, central
Body Segment: Hand/wrist
Bone Type: Short bone (carpal)
Shape: Largest carpal; head articulates with lunate/scaphoid
Surface Anatomy: Deep in central palm
Anatomical Landmarks: Head, waist, body
Articulations: Scaphoid, lunate, hamate, trapezoid, metacarpal III primarily
Muscle Attachments: None direct
Ligament Attachments: Intercarpal ligaments; capitometacarpal
Tendon Attachments: None direct
Blood Supply: Palmar/dorsal carpal branches
Venous Drainage: Carpal venous plexus
Ossification: First carpal to ossify (~1 year)
Biomechanical Function: Central pillar distal carpal row; load from lunate to third MC
Clinical Importance: Fractures uncommon; scaphocapitate syndrome in high-energy injury
Common Fractures: Body/head fractures rare — high energy
Healing Time: 8–12 weeks if uncomplicated
Imaging Recommendations: CT for occult fracture; wrist series
References: see PubMed [capitate fracture]

RECORD: Triquetrum

ID: BONE-UE-009
Name: Triquetrum
Latin Name: Os triquetrum
Alternative Names: Pyramidal bone, cuneiform (historical)
Region: Wrist — proximal carpal row, ulnar side
Body Segment: Hand/wrist
Bone Type: Short bone (carpal)

Shape: Pyramidal; oval facet for pisiform
Surface Anatomy: Ulnar wrist deep to FCU
Anatomical Landmarks: Dorsal ridge often fracture site
Articulations: Lunate, hamate, pisiform
Muscle Attachments: None direct
Ligament Attachments: LT ligament; TFCC; pisohamate/pisometacarpal via pisiform
Tendon Attachments: None direct
Blood Supply: Ulnar artery branches
Venous Drainage: Ulnar side carpal plexus
Ossification: Ossifies ~3 years
Biomechanical Function: Ulnar carpal stability; extension of TFCC load sharing
Clinical Importance: Dorsal chip fracture common after FOOSH wrist extension/ulnar deviation
Common Fractures: Dorsal avulsion chip (most common carpal fracture after scaphoid)
Healing Time: 4–6 weeks typically conservative
Imaging Recommendations: Lateral wrist; CT if occult pain persists
References: see PubMed [triquetrum dorsal fracture]

RECORD: Pisiform

ID: BONE-UE-010
Name: Pisiform
Latin Name: Os pisiforme
Alternative Names: Pea-shaped bone
Region: Wrist — proximal carpal row, ulnar palmar
Body Segment: Hand/wrist
Bone Type: Sesamoid bone (in FCU tendon)
Shape: Pea-shaped
Surface Anatomy: Palpable ulnar wrist in flexion
Anatomical Landmarks: Single articular facet for triquetrum
Articulations: Triquetrum
Muscle Attachments: FCU tendon encases pisiform
Ligament Attachments: Pisohamate, pisometacarpal
Tendon Attachments: FCU insertion via pisiform as sesamoid
Blood Supply: Ulnar artery branches
Venous Drainage: Ulnar venous plexus
Ossification: Ossifies ~9–12 years
Biomechanical Function: Pulley/increase moment arm for FCU; ulnar wrist stability
Clinical Importance: Fractures rare; pisotriquetral arthritis; hook of hamate nearby in cyclists/golfers
Common Fractures: Direct blow fractures uncommon
Healing Time: 6–8 weeks
Imaging Recommendations: Carpal tunnel view; CT for pisotriquetral
References: see PubMed [pisiform fracture FCU]

RECORD: Trapezium

ID: BONE-UE-011

Name: Trapezium

Latin Name: Os trapezium

Alternative Names: Greater multangular

Region: Wrist — distal carpal row, radial

Body Segment: Hand/wrist

Bone Type: Short bone (carpal)

Shape: Irregular; saddle-shaped facet for thumb MC

Surface Anatomy: Thenar eminence base

Anatomical Landmarks: Saddle facet, tubercle

Articulations: Scaphoid, trapezoid, 1st metacarpal

Muscle Attachments: Thenar muscles attach to thumb MC not trapezium directly

Ligament Attachments: Beak ligament (trapeziometacarpal); scaphotrapezial

Tendon Attachments: APL/EPB cross nearby

Blood Supply: Radial artery branches

Venous Drainage: Radial venous plexus

Ossification: Ossifies ~4 years

Biomechanical Function: Saddle joint enables thumb opposition — key to grip

Clinical Importance: 1st CMC OA most common hand arthritis; Bennett/Rolando at MC base

Common Fractures: Rolando/trapezial body rare

Healing Time: 8–12 weeks

Imaging Recommendations: Robert view for CMC; MRI for ligament

References: Eaton CMC OA staging; see PubMed [trapeziometacarpal osteoarthritis]

RECORD: Trapezoid

ID: BONE-UE-012

Name: Trapezoid

Latin Name: Os trapezoideum

Alternative Names: Lesser multangular

Region: Wrist — distal carpal row

Body Segment: Hand/wrist

Bone Type: Short bone (carpal)

Shape: Smallest carpal in distal row; wedge-shaped

Surface Anatomy: Deep

Anatomical Landmarks: Four articular facets

Articulations: Scaphoid, capitate, trapezium, 2nd metacarpal

Muscle Attachments: None direct

Ligament Attachments: Intercarpal ligaments

Tendon Attachments: None direct

Blood Supply: Radial artery branches

Venous Drainage: Carpal plexus

Ossification: Ossifies ~6 years

Biomechanical Function: Stabilizes index ray; keystone between scaphoid and 2nd MC

Clinical Importance: Isolated fracture extremely rare

Common Fractures: Rare high-energy

Healing Time: 8 weeks if uncomplicated

Imaging Recommendations: CT if suspected

References: see PubMed [trapezoid fracture]

RECORD: Hamate

ID: BONE-UE-013

Name: Hamate

Latin Name: Os hamatum

Alternative Names: Unciform bone

Region: Wrist — distal carpal row, ulnar

Body Segment: Hand/wrist

Bone Type: Short bone (carpal)

Shape: Hook process palmarly — critical landmark

Surface Anatomy: Hook palpable deep to hypothenar; Guyon's canal lateral to hook

Anatomical Landmarks: Hook (hamulus), body, articular facets for 4th/5th MC

Articulations: Triquetrum, lunate, capitate, 4th/5th metacarpals

Muscle Attachments: Flexor digiti minimi, opponens digiti minimi (via hook ligaments)

Ligament Attachments: Pisohamate; transverse carpal (flexor retinaculum) attaches hook

Tendon Attachments: Flexor tendons pass medial to hook

Blood Supply: Ulnar artery branches

Venous Drainage: Ulnar venous plexus

Ossification: Hook ossifies ~12–15 years

Biomechanical Function: Ulnar carpal stability; flexor pulley anchor

Clinical Importance: Hook fracture in racket/cycling/golf; ulnar nerve in Guyon's canal

Common Fractures: Hook avulsion/stress fracture

Healing Time: 6–12 weeks; non-union may need excision

Imaging Recommendations: Carpal tunnel view, CT (hook often missed on AP)

References: see PubMed [hamate hook fracture cyclists]

RECORD: Metacarpals (Group)

ID: BONE-UE-014

Name: Metacarpals (Group)

Latin Name: Ossa metacarpi I–V

Alternative Names: MC bones

Region: Hand

Body Segment: Hand

Bone Type: Long bones (five)

Shape: Base-shaft-head; 2nd/3rd most fixed; 4th/5th more mobile

Surface Anatomy: Dorsal shafts palpable; knuckles = heads at MCP

Anatomical Landmarks: Base, shaft, head, neck (Boxer's fracture 5th neck common)

Articulations: Carpals proximally, proximal phalanges distally, intermetacarpal

Muscle Attachments: Dorsal interossei, palmar interossei, lumbricals, thenar/hypothenar, adductor pollicis

Ligament Attachments: Deep transverse metacarpal; collateral at MCP

Tendon Attachments: Flexor/extensor tendons pass to digits

Blood Supply: Dorsal/palmar metacarpal arteries from deep palmar arch
Venous Drainage: Deep palmar venous plexus
Ossification: Epiphyses fuse ~16–18 years
Biomechanical Function: Transverse and longitudinal arches of hand; power grip platform
Clinical Importance: Boxer's fracture (5th MC neck); Bennett/ Rolando 1st MC base
Common Fractures: Neck (5th), shaft, base intra-articular (Bennett)
Healing Time: 4–6 weeks; angulation acceptable 5th MC ~40°
Imaging Recommendations: Hand PA/oblique/lateral; CT for base fractures
References: see PubMed [metacarpal neck fracture management]

RECORD: Phalanges (Group)

ID: BONE-UE-015
Name: Phalanges (Group)
Latin Name: Ossa phalanges (proximal, media, distales)
Alternative Names: Finger bones
Region: Digits
Body Segment: Hand — digits
Bone Type: Long bones (14 total per hand)
Shape: Proximal (5), middle (4), distal (4); thumb lacks middle
Surface Anatomy: Interphalangeal joints palpable dorsally
Anatomical Landmarks: Base, shaft, head, tuft (distal)
Articulations: MCP, PIP, DIP; thumb IP
Muscle Attachments: Lumbricals, interossei (via extensor mechanism)
Ligament Attachments: Collateral ligaments volar/dorsal plates
Tendon Attachments: FDS/FDP flexors; central slip and lateral bands extensors
Blood Supply: Digital arteries from superficial/deep palmar arches
Venous Drainage: Digital veins
Ossification: Epiphyses fuse ~16–18 years
Biomechanical Function: Precision and power grip; tendon pulley system
Clinical Importance: Mallet (extensor avulsion); jersey (FDP); trigger finger (A1 pulley)
Common Fractures: Tuft crush, shaft, intra-articular PIP (pilon)
Healing Time: 3–6 weeks phalangeal; tendon injuries 6–12+ weeks
Imaging Recommendations: Finger lateral for mallet; MRI for occult jersey in children
References: see PubMed [phalangeal fracture buddy taping]

2. Muscles

RECORD: Deltoid

ID: MUS-UE-001

Name: Deltoid

Latin Name: Musculus deltoideus

Region: Shoulder

Compartment: Superficial shoulder

Origin: Lateral third clavicle, acromion, spine of scapula

Insertion: Deltoid tuberosity humerus

Innervation: Axillary nerve

Nerve Roots: C5–C6

Blood Supply: Posterior circumflex humeral artery

Fascial Connections: Thoracolumbar fascia continuity via trapezius; clavipectoral fascia

Fiber Direction: Multipennate; anterior oblique, middle vertical, posterior oblique

Pennation: Multipennate

Primary Action: Shoulder abduction (middle), flexion (ant), extension/ER (post)

Secondary Actions: Horizontal abduction (post), scapular elevation assist

Stabilizing Function: GH joint compression in mid-range

Dynamic Function: Primary power for arm elevation and overhead activity

Role During Gait: N/A — UE swing phase assist

Role During Throwing: Deceleration and follow-through control; excessive use → impingement

Role During Lifting: Overhead press/lateral raise prime mover

Synergists: Supraspinatus, trapezius

Antagonists: Latissimus dorsi, pectoralis major

Length-Tension Relationship: Optimal tension 0–60° abduction

EMG Data: High EMG in abduction 90–120°; throwing deceleration

Stretch Position: Adduction + IR

Strength Position: 90° abduction scapular plane

Common Trigger Points: Mid-fiber trigger common

Pain Referral Pattern: Lateral shoulder referral

Palpation: Palpate over acromion edges and muscle belly

Manual Muscle Test: Abduction MMT; isolate anterior/posterior with rotation

Clinical Importance: Impingement-related deltoid dominance

Common Injuries: Strain, contusion

Healing Timeline: 2–6 weeks

Contraindications: Avoid painful arc loading early post rotator cuff repair

References: Kendall MMT; see PubMed [deltoid EMG throwing]

RECORD: Supraspinatus

ID: MUS-UE-002

Name: Supraspinatus

Latin Name: Musculus supraspinatus

Region: Shoulder

Compartment: Rotator cuff — supraspinous fossa

Origin: Supraspinous fossa scapula

Insertion: Superior facet greater tubercle humerus

Innervation: Suprascapular nerve

Nerve Roots: C5–C6

Blood Supply: Suprascapular artery

Fascial Connections: Rotator cuff interval; coracohumeral ligament region

Fiber Direction: Oblique

Pennation: Unipennate

Primary Action: Initiates abduction; stabilizes humeral head

Secondary Actions: External rotation assist minor

Stabilizing Function: Prevents superior humeral migration

Dynamic Function: Force couple with infraspinatus/subscapularis

Role During Gait: N/A

Role During Throwing: Critical in cocking and early acceleration (GH compression)

Role During Lifting: Initiates lifting arm from side

Synergists: Deltoid, infraspinatus

Antagonists: Latissimus, pec major

Length-Tension Relationship: Weakened when shortened under cuff tear

EMG Data: Active 0–30° abduction; full can position high

Stretch Position: Adduction + extension

Strength Position: Full can 90° scapular plane

Common Trigger Points: Supraspinatus fossa deep

Pain Referral Pattern: Lateral deltoid referral

Palpation: Deep to trapezius superior scapula

Manual Muscle Test: Full/empty can test

Clinical Importance: Most commonly torn RC muscle; subacromial impingement

Common Injuries: Tendinopathy, partial/full tear

Healing Timeline: Tendinopathy 6–12 wk; tear surgical 4–6 mo RTS

Contraindications: Avoid heavy overhead early post repair

References: Jobe empty can; see PubMed [supraspinatus tear incidence]

RECORD: Infraspinatus

ID: MUS-UE-003

Name: Infraspinatus

Latin Name: Musculus infraspinatus

Region: Shoulder

Compartment: Rotator cuff — infraspinous fossa

Origin: Infraspinous fossa

Insertion: Middle facet greater tubercle

Innervation: Suprascapular nerve

Nerve Roots: C5–C6

Blood Supply: Suprascapular and circumflex scapular

Fascial Connections: Thoracolumbar fascia via teres minor
Fiber Direction: Oblique/multipennate
Pennation: Multipennate
Primary Action: External rotation
Secondary Actions: Horizontal extension, GH stability
Stabilizing Function: Posterior force couple
Dynamic Function: Deceleration ER in throwing
Role During Gait: N/A
Role During Throwing: Major ER during cocking; eccentric control in follow-through
Role During Lifting: External rotation load in pulling
Synergists: Teres minor, posterior deltoid
Antagonists: Subscapularis, latissimus, pec major
Length-Tension Relationship: Weakness with anterior tightness
EMG Data: High in throwing deceleration phase
Stretch Position: Flexion + IR
Strength Position: 90° abduction full ER
Common Trigger Points: Mid-belly infraspinous fossa
Pain Referral Pattern: Posterior shoulder/scapula
Palpation: Deep pressure infraspinous fossa
Manual Muscle Test: ER MMT side-lying
Clinical Importance: Posterior cuff tear → ER weakness
Common Injuries: Tear, tendinopathy
Healing Timeline: 6–12 weeks tendinopathy
Contraindications: Post-op ER precautions per protocol
References: see PubMed [infraspinatus tear ER strength]

RECORD: Teres minor

ID: MUS-UE-004
Name: Teres minor
Latin Name: Musculus teres minor
Region: Shoulder
Compartment: Rotator cuff
Origin: Lateral border scapula
Insertion: Inferior facet greater tubercle
Innervation: Axillary nerve
Nerve Roots: C5
Blood Supply: Circumflex scapular
Fascial Connections: Blends with infraspinatus
Fiber Direction: Oblique
Pennation: Unipennate
Primary Action: External rotation
Secondary Actions: Adduction assist, GH stability
Stabilizing Function: Posterior cuff force couple
Dynamic Function: Fine-tunes humeral head centering

Role During Gait: N/A
Role During Throwing: ER in late cocking
Role During Lifting: Pulling/lifting stability
Synergists: Infraspinatus
Antagonists: Subscapularis
Length-Tension Relationship: Often intact when IS torn
EMG Data: Moderate in ER
Stretch Position: Adduction + IR
Strength Position: Side-lying ER
Common Trigger Points: Rare isolated trigger
Pain Referral Pattern: Posterior shoulder
Palpation: Inferior scapular border
Manual Muscle Test: ER MMT
Clinical Importance: Isolated tear uncommon
Common Injuries: Tendinopathy rare
Healing Timeline: 6–8 weeks
Contraindications: Same as infraspinatus post-op
References: see PubMed [teres minor isolated tear]

RECORD: Subscapularis

ID: MUS-UE-005
Name: Subscapularis
Latin Name: Musculus subscapularis
Region: Shoulder
Compartment: Rotator cuff — subscapular fossa
Origin: Subscapular fossa
Insertion: Lesser tubercle humerus
Innervation: Upper/Lower subscapular nerves
Nerve Roots: C5–C6
Blood Supply: Subscapular artery branches
Fascial Connections: Axillary fold; coracoid region
Fiber Direction: Multipennate
Pennation: Multipennate
Primary Action: Internal rotation
Secondary Actions: Adduction, GH anterior stability
Stabilizing Function: Prevents anterior translation
Dynamic Function: Lift-off and belly-press function
Role During Gait: N/A
Role During Throwing: IR during acceleration phase
Role During Lifting: IR in lifting/carrying
Synergists: Pectoralis major, latissimus
Antagonists: Infraspinatus, teres minor
Length-Tension Relationship: Tight subscap → IR loss
EMG Data: High in IR and push activities

Stretch Position: Abduction + ER
Strength Position: IR at 0° or 90°
Common Trigger Points: Deep subscapular fossa
Pain Referral Pattern: Anterior shoulder
Palpation: Deep to pec major medial
Manual Muscle Test: Lift-off, belly-press, bear hug
Clinical Importance: Hidden lesion in anterior instability
Common Injuries: Tear (traumatic), tendinopathy
Healing Timeline: 8–12 weeks; large tear surgical
Contraindications: Protect IR post anterior repair
References: see PubMed [subscapularis tear lift-off]

RECORD: Biceps brachii

ID: MUS-UE-006
Name: Biceps brachii
Latin Name: Musculus biceps brachii
Region: Arm
Compartment: Anterior arm
Origin: Long head supraglenoid; short head coracoid
Insertion: Radial tuberosity; bicipital aponeurosis
Innervation: Musculocutaneous nerve
Nerve Roots: C5–C6
Blood Supply: Brachial artery
Fascial Connections: Antebrachial fascia via aponeurosis
Fiber Direction: Long head parallel; short head fusiform
Pennation: Fusiform
Primary Action: Elbow flexion, forearm supination
Secondary Actions: Shoulder flexion (long head)
Stabilizing Function: GH anterior stabilizer (long head)
Dynamic Function: Dynamic elbow flexor
Role During Gait: N/A
Role During Throwing: Elbow flexion in throwing; long head stress in cocking
Role During Lifting: Lifting/curling
Synergists: Brachialis, brachioradialis
Antagonists: Triceps
Length-Tension Relationship: Two-joint muscle — length tension varies with shoulder position
EMG Data: High in supinated flexion
Stretch Position: Elbow extension + pronation
Strength Position: Supinated flexion 90°
Common Trigger Points: Mid-belly anterior arm
Pain Referral Pattern: Anterior arm/forearm
Palpation: Anterior arm
Manual Muscle Test: Flexion/supination MMT
Clinical Importance: SLAP lesions affect long head

Common Injuries: Tendinopathy, partial tear, rupture
Healing Timeline: Tendinopathy 6–12 wk
Contraindications: Avoid aggressive curl post SLAP repair
References: see PubMed [biceps tendinopathy treatment]

RECORD: Brachialis

ID: MUS-UE-007
Name: Brachialis
Latin Name: Musculus brachialis
Region: Arm
Compartment: Anterior arm deep
Origin: Distal half anterior humerus
Insertion: Ulnar tuberosity
Innervation: Musculocutaneous (primary), radial (minor)
Nerve Roots: C5–C6
Blood Supply: Brachial artery
Fascial Connections: Deep to biceps
Fiber Direction: Oblique
Pennation: Unipennate
Primary Action: Elbow flexion (all positions)
Secondary Actions: Minimal shoulder action
Stabilizing Function: Primary flexor regardless of forearm position
Dynamic Function: Power flexion
Role During Gait: N/A
Role During Throwing: Elbow flexion acceleration
Role During Lifting: Primary lifting flexor
Synergists: Biceps, brachioradialis
Antagonists: Triceps
Length-Tension Relationship: Strongest elbow flexor mechanically
EMG Data: Highest flexion EMG neutral forearm
Stretch Position: Full elbow extension
Strength Position: Flexion neutral forearm
Common Trigger Points: Deep to biceps distal arm
Pain Referral Pattern: Forearm
Palpation: Deep anterior humerus
Manual Muscle Test: Elbow flexion MMT neutral
Clinical Importance: Often spared in biceps rupture
Common Injuries: Strain uncommon
Healing Timeline: 4–6 weeks
Contraindications: Standard flexion rehab
References: see PubMed [brachialis EMG flexion]

RECORD: Triceps brachii

ID: MUS-UE-008

Name: Triceps brachii
Latin Name: Musculus triceps brachii
Region: Arm
Compartment: Posterior arm
Origin: Long head infraglenoid; lateral/medial heads posterior humerus
Insertion: Olecranon process
Innervation: Radial nerve
Nerve Roots: C6–C8
Blood Supply: Deep brachial artery
Fascial Connections: Fascia lata equivalent — lateral intermuscular septum
Fiber Direction: Long head parallel; lateral/medial multipennate
Pennation: Multipennate
Primary Action: Elbow extension
Secondary Actions: Shoulder extension (long head), adduction
Stabilizing Function: Elbow stability in push/weight-bearing
Dynamic Function: Deceleration control in falling onto hand
Role During Gait: N/A
Role During Throwing: Extension in push-off from hand
Role During Lifting: Pressing, pushing, bench press
Synergists: Anconeus
Antagonists: Biceps, brachialis
Length-Tension Relationship: Long head crosses shoulder
EMG Data: High in push-ups, throwing follow-through
Stretch Position: Full flexion
Strength Position: Extension from 90° flexion
Common Trigger Points: Mid-belly posterior arm
Pain Referral Pattern: Posterior arm/elbow
Palpation: Posterior arm
Manual Muscle Test: Extension MMT
Clinical Importance: Tendinopathy rare; rupture post steroid
Common Injuries: Strain, avulsion
Healing Timeline: 6–10 weeks
Contraindications: Protect extension post olecranon ORIF
References: see PubMed [triceps tendinopathy]

RECORD: Brachioradialis

ID: MUS-UE-009
Name: Brachioradialis
Latin Name: Musculus brachioradialis
Region: Forearm
Compartment: Lateral forearm superficial
Origin: Lateral supracondylar ridge humerus
Insertion: Styloid process radius
Innervation: Radial nerve

Nerve Roots: C5–C6
Blood Supply: Radial recurrent artery
Fascial Connections: Mobile wad of Henry
Fiber Direction: Oblique
Pennation: Fusiform
Primary Action: Elbow flexion (especially mid-pronation)
Secondary Actions: Forearm stabilization
Stabilizing Function: Stabilizes elbow during grip
Dynamic Function: Adaptive flexor when other flexors injured
Role During Gait: N/A
Role During Throwing: Elbow positioning in throw
Role During Lifting: Hammer curl, lifting
Synergists: Brachialis, biceps
Antagonists: Triceps
Length-Tension Relationship: Most active flexion in neutral/pronation
EMG Data: Moderate-high hammer curl EMG
Stretch Position: Elbow extension + wrist flexion
Strength Position: Mid-pronation flexion
Common Trigger Points: Proximal lateral forearm
Pain Referral Pattern: Lateral forearm/elbow
Palpation: Lateral forearm
Manual Muscle Test: Flexion mid-pronation MMT
Clinical Importance: Used as tendon graft donor
Common Injuries: Strain
Healing Timeline: 4–6 weeks
Contraindications: None specific
References: see PubMed [brachioradialis EMG]

RECORD: Pronator teres

ID: MUS-UE-010
Name: Pronator teres
Latin Name: Musculus pronator teres
Region: Forearm
Compartment: Anterior superficial
Origin: Medial epicondyle + coronoid
Insertion: Lateral mid-radius
Innervation: Median nerve
Nerve Roots: C6–C7
Blood Supply: Ulnar/radial arteries
Fascial Connections: Common flexor origin
Fiber Direction: Oblique
Pennation: Unipennate
Primary Action: Forearm pronation
Secondary Actions: Elbow flexion assist

Stabilizing Function: Pronation power
Dynamic Function: Grip setup pronation
Role During Gait: N/A
Role During Throwing: Pronation in throwing change-up grip
Role During Lifting: Pronated lifting
Synergists: Pronator quadratus, FCR
Antagonists: Supinator, biceps
Length-Tension Relationship: Active in resisted pronation
EMG Data: High resisted pronation EMG
Stretch Position: Supination + extension
Strength Position: Pronation elbow extended
Common Trigger Points: Proximal volar forearm
Pain Referral Pattern: Volar forearm/elbow
Palpation: Volar proximal forearm
Manual Muscle Test: Resisted pronation
Clinical Importance: Pronator syndrome (median)
Common Injuries: Strain, median nerve irritation
Healing Timeline: 4–8 weeks
Contraindications: Avoid aggressive pronation with median irritation
References: see PubMed [pronator teres median nerve]

RECORD: Flexor carpi radialis (FCR)

ID: MUS-UE-011
Name: Flexor carpi radialis (FCR)
Latin Name: Musculus flexor carpi radialis
Region: Forearm
Compartment: Anterior superficial
Origin: Medial epicondyle
Insertion: Base 2nd/3rd metacarpals
Innervation: Median nerve
Nerve Roots: C6–C7
Blood Supply: Ulnar artery branches
Fascial Connections: Common flexor origin
Fiber Direction: Parallel
Pennation: Unipennate
Primary Action: Wrist flexion, radial deviation
Secondary Actions: Elbow flexion assist minor
Stabilizing Function: Radial wrist stability
Dynamic Function: Grip force transmission
Role During Gait: N/A
Role During Throwing: Wrist flexion in racquet sports
Role During Lifting: Wrist flexion in lifting
Synergists: FCU, palmaris longus
Antagonists: ECRL, ECRB

Length-Tension Relationship: Active in grip
EMG Data: Moderate grip EMG
Stretch Position: Extension + ulnar deviation
Strength Position: Flexion + radial deviation
Common Trigger Points: Proximal volar forearm
Pain Referral Pattern: Volar wrist/thumb
Palpation: Medial forearm
Manual Muscle Test: Wrist flexion/radial deviation MMT
Clinical Importance: Part of golfer's elbow complex
Common Injuries: Tendinopathy
Healing Timeline: 6–12 weeks
Contraindications: Avoid aggravating flexion in medial epicondylitis
References: see PubMed [FCR tendinopathy]

RECORD: Flexor carpi ulnaris (FCU)

ID: MUS-UE-012
Name: Flexor carpi ulnaris (FCU)
Latin Name: Musculus flexor carpi ulnaris
Region: Forearm
Compartment: Anterior/medial
Origin: Medial epicondyle + olecranon
Insertion: Pisiform, hook hamate, 5th MC base
Innervation: Ulnar nerve (motor)
Nerve Roots: C7–T1
Blood Supply: Ulnar artery
Fascial Connections: Common flexor + ulnar side
Fiber Direction: Parallel
Pennation: Unipennate
Primary Action: Wrist flexion, ulnar deviation
Secondary Actions: Elbow flexion assist
Stabilizing Function: Ulnar wrist stability in grip/power
Dynamic Function: Stabilizes ulnar side wrist
Role During Gait: N/A
Role During Throwing: Ulnar deviation in golf/tennis
Role During Lifting: Heavy grip flexion
Synergists: FCR, digitorum flexors
Antagonists: ECU, ECRL
Length-Tension Relationship: High in ulnar deviation tasks
EMG Data: Moderate-high
Stretch Position: Extension + radial deviation
Strength Position: Flexion + ulnar deviation
Common Trigger Points: Pisiform region
Pain Referral Pattern: Ulnar wrist/hand
Palpation: Medial forearm/pisiform

Manual Muscle Test: Flexion/ulnar deviation MMT
Clinical Importance: Golfer's elbow; FCU strength test for ulnar nerve
Common Injuries: Tendinopathy, strain
Healing Timeline: 6–12 weeks
Contraindications: Medial epicondylitis precautions
References: see PubMed [FCU golfer elbow]

RECORD: Flexor digitorum superficialis (FDS)

ID: MUS-UE-013
Name: Flexor digitorum superficialis (FDS)
Latin Name: Musculus flexor digitorum superficialis
Region: Forearm
Compartment: Anterior intermediate
Origin: Medial epicondyle, coronoid, radius
Insertion: Middle phalanges 2–5
Innervation: Median nerve (anterior interosseous branch to index)
Nerve Roots: C7–T1
Blood Supply: Ulnar/radial arteries
Fascial Connections: Flexor digitorum profundus sheath
Fiber Direction: Oblique to digits
Pennation: Unipennate per digit
Primary Action: Flexion PIP (and MCP assist)
Secondary Actions: Wrist flexion assist
Stabilizing Function: Finger flexion independent per digit
Dynamic Function: Grip closure
Role During Gait: N/A
Role During Throwing: Ball grip in throwing
Role During Lifting: Grasping objects
Synergists: FDP, lumbricals
Antagonists: EDC, EDC
Length-Tension Relationship: FDS blocks FDP when PIP held
EMG Data: High in grip
Stretch Position: Digit extension
Strength Position: Resisted PIP flexion isolated
Common Trigger Points: Mid-forearm
Pain Referral Pattern: Palmar digits
Palpation: Individual digit FDS test (hold FDP out)
Manual Muscle Test: Isolated FDS test per digit
Clinical Importance: Trigger finger differential; A1 pulley
Common Injuries: Tendinopathy, laceration
Healing Timeline: 8–12 weeks post repair
Contraindications: Protect flexor tendon repair
References: see PubMed [flexor tendon rehabilitation Kleinert]

RECORD: Flexor digitorum profundus (FDP)

ID: MUS-UE-014

Name: Flexor digitorum profundus (FDP)

Latin Name: Musculus flexor digitorum profundus

Region: Forearm

Compartment: Anterior deep

Origin: Proximal ulna/interosseous membrane

Insertion: Distal phalanges 2–5

Innervation: Median (2–3), Ulnar (4–5) — anterior interosseous/median/ulnar

Nerve Roots: C8–T1

Blood Supply: Anterior interosseous artery

Fascial Connections: Shared synovial sheath

Fiber Direction: Long tendons

Pennation: Unipennate

Primary Action: Flexion DIP (and PIP/MCP)

Secondary Actions: Wrist flexion assist

Stabilizing Function: Terminal grip strength

Dynamic Function: Power grip

Role During Gait: N/A

Role During Throwing: Finger flexion gripping ball

Role During Lifting: Heavy grip/lifting

Synergists: FDS, lumbricals

Antagonists: EDC

Length-Tension Relationship: Critical for tip pinch strength

EMG Data: Maximal in tip grip

Stretch Position: DIP extension

Strength Position: Resisted DIP flexion

Common Trigger Points: Deep forearm

Pain Referral Pattern: Distal phalanx pain

Palpation: Distal phalanx

Manual Muscle Test: DIP flexion MMT

Clinical Importance: Jersey finger (ring finger common)

Common Injuries: Avulsion, laceration

Healing Timeline: 8–12+ weeks; Zone I 6–8 wk conservative

Contraindications: No resisted flexion early post repair

References: see PubMed [jersey finger FDP]

RECORD: Extensor carpi radialis longus (ECRL)

ID: MUS-UE-015

Name: Extensor carpi radialis longus (ECRL)

Latin Name: Musculus extensor carpi radialis longus

Region: Forearm

Compartment: Posterior superficial mobile wad

Origin: Lateral supracondylar ridge

Insertion: Base 2nd metacarpal
Innervation: Radial nerve
Nerve Roots: C6
Blood Supply: Radial recurrent
Fascial Connections: Mobile wad
Fiber Direction: Parallel
Pennation: Unipennate
Primary Action: Wrist extension, radial deviation
Secondary Actions: Elbow flexion assist minor
Stabilizing Function: Radial wrist stability
Dynamic Function: Grip extension balance
Role During Gait: N/A
Role During Throwing: Wrist extension in backhand
Role During Lifting: Lifting with extended wrist
Synergists: ECRB, ECRB
Antagonists: FCR, FCU
Length-Tension Relationship: Active early in wrist extension
EMG Data: Moderate-high extension
Stretch Position: Flexion + ulnar deviation
Strength Position: Extension + radial deviation
Common Trigger Points: Proximal dorsal forearm
Pain Referral Pattern: Dorsal wrist
Palpation: Dorsal forearm
Manual Muscle Test: Extension/radial deviation MMT
Clinical Importance: Often spared in radial nerve palsy vs ECRB
Common Injuries: Strain
Healing Timeline: 4–6 weeks
Contraindications: None
References: see PubMed [ECRL wrist extension]

RECORD: Extensor carpi radialis brevis (ECRB)

ID: MUS-UE-016
Name: Extensor carpi radialis brevis (ECRB)
Latin Name: Musculus extensor carpi radialis brevis
Region: Forearm
Compartment: Posterior superficial
Origin: Lateral epicondyle (common extensor)
Insertion: Base 3rd metacarpal
Innervation: Radial nerve (PIN)
Nerve Roots: C6–C7
Blood Supply: Radial recurrent
Fascial Connections: Common extensor tendon
Fiber Direction: Oblique
Pennation: Unipennate

Primary Action: Wrist extension

Secondary Actions: Radial deviation assist

Stabilizing Function: Primary tennis elbow tendon origin

Dynamic Function: Stabilizes wrist in grip

Role During Gait: N/A

Role During Throwing: Wrist extension in forehand/serve

Role During Lifting: Grip with extended wrist

Synergists: ECRL, ECU

Antagonists: FCR, FCU

Length-Tension Relationship: Degenerative tendinopathy common at origin

EMG Data: High in gripping with extended wrist

Stretch Position: Flexion

Strength Position: Extension neutral deviation

Common Trigger Points: Common extensor origin lateral epicondyle

Pain Referral Pattern: Lateral epicondyle → dorsum forearm

Palpation: Lateral epicondyle

Manual Muscle Test: Resisted wrist extension middle finger (Cozen)

Clinical Importance: Primary pathology in lateral epicondylalgia

Common Injuries: Tendinopathy, tear rare

Healing Timeline: 6–12 months chronic

Contraindications: Avoid painful gripping early

References: see PubMed [ECRB tennis elbow Cochrane]

RECORD: Extensor carpi ulnaris (ECU)

ID: MUS-UE-017

Name: Extensor carpi ulnaris (ECU)

Latin Name: Musculus extensor carpi ulnaris

Region: Forearm

Compartment: Posterior ulnar

Origin: Lateral epicondyle + posterior ulna

Insertion: Base 5th metacarpal

Innervation: Ulnar nerve branch (PIN via radial) — actually PIN C7–C8

Nerve Roots: C7–C8

Blood Supply: Ulnar artery branches

Fascial Connections: Sixth extensor compartment

Fiber Direction: Oblique

Pennation: Unipennate

Primary Action: Wrist extension, ulnar deviation

Secondary Actions: Forearm stabilization

Stabilizing Function: Ulnar wrist stability — critical in tennis/golf

Dynamic Function: ECU synergy in ulnar deviation

Role During Gait: N/A

Role During Throwing: Ulnar deviation in backhand/forehand

Role During Lifting: Hammer/tool use

Synergists: ECRL, ECRB
Antagonists: FCR, FCU
Length-Tension Relationship: ECU subsheath stabilizes in supination
EMG Data: High in ulnar deviation sports
Stretch Position: Flexion + radial deviation
Strength Position: Extension + ulnar deviation
Common Trigger Points: Ulnar wrist dorsally
Pain Referral Pattern: Ulnar dorsal wrist
Palpation: Ulnar snuffbox region
Manual Muscle Test: Extension/ulnar deviation MMT
Clinical Importance: ECU tendinopathy/subluxation
Common Injuries: Tendinopathy, instability
Healing Timeline: 6–12 weeks
Contraindications: Immobilize in tennis/golf ECU injury
References: see PubMed [ECU subluxation tennis]

RECORD: Extensor digitorum communis (EDC)

ID: MUS-UE-018
Name: Extensor digitorum communis (EDC)
Latin Name: Musculus extensor digitorum
Region: Forearm
Compartment: Posterior superficial
Origin: Common extensor origin
Insertion: Extensor expansions digits 2–5
Innervation: Posterior interosseous nerve
Nerve Roots: C7–C8
Blood Supply: Posterior interosseous artery
Fascial Connections: Extensor hood
Fiber Direction: Multiple tendons
Pennation: Unipennate
Primary Action: Extension MCP, PIP, DIP via hood
Secondary Actions: Wrist extension assist
Stabilizing Function: Finger extension for release/grip opening
Dynamic Function: Fine motor extension
Role During Gait: N/A
Role During Throwing: Finger extension in release phase throw
Role During Lifting: Opening hand from grip
Synergists: Extensor indicis, EDM
Antagonists: FDS, FDP
Length-Tension Relationship: Complex extensor mechanism
EMG Data: Moderate extension activities
Stretch Position: Finger flexion
Strength Position: MCP extension
Common Trigger Points: Dorsal forearm

Pain Referral Pattern: Dorsum hand
Palpation: Dorsal forearm
Manual Muscle Test: Finger extension MMT
Clinical Importance: PIN palsy → finger drop
Common Injuries: Laceration, sagittal band rupture
Healing Timeline: 8–12 weeks post repair
Contraindications: Protect extensor repair
References: see PubMed [extensor tendon repair]

RECORD: Abductor pollicis longus (APL)

ID: MUS-UE-019
Name: Abductor pollicis longus (APL)
Latin Name: Musculus abductor pollicis longus
Region: Forearm
Compartment: Posterior radial
Origin: Ulna/radius posterior
Insertion: Base 1st metacarpal
Innervation: Posterior interosseous nerve
Nerve Roots: C7–C8
Blood Supply: Posterior interosseous artery
Fascial Connections: First dorsal compartment
Fiber Direction: Oblique
Pennation: Unipennate
Primary Action: Thumb abduction, extension assist
Secondary Actions: Radial wrist abduction
Stabilizing Function: De Quervain compartment
Dynamic Function: Thumb positioning
Role During Gait: N/A
Role During Throwing: Thumb abduction in grip
Role During Lifting: Pinch/lifting
Synergists: EPB
Antagonists: Adductor pollicis
Length-Tension Relationship: First dorsal compartment stenosis
EMG Data: Moderate in pinch
Stretch Position: Adduction + flexion
Strength Position: Abduction/extension
Common Trigger Points: Anatomic snuffbox radial
Pain Referral Pattern: Radial wrist/thumb
Palpation: Snuffbox
Manual Muscle Test: Thumb abduction MMT
Clinical Importance: De Quervain tenosynovitis
Common Injuries: Tenosynovitis
Healing Timeline: 4–8 weeks
Contraindications: Avoid thumb repetitive abduction

References: see PubMed [De Quervain APL EPB]

RECORD: Extensor pollicis brevis (EPB)

ID: MUS-UE-020

Name: Extensor pollicis brevis (EPB)

Latin Name: Musculus extensor pollicis brevis

Region: Forearm

Compartment: Posterior radial

Origin: Radius/ulna posterior

Insertion: Proximal phalanx thumb

Innervation: Posterior interosseous nerve

Nerve Roots: C7–C8

Blood Supply: Posterior interosseous artery

Fascial Connections: First dorsal compartment with APL

Fiber Direction: Oblique

Pennation: Unipennate

Primary Action: Thumb extension MCP

Secondary Actions: Radial wrist extension assist

Stabilizing Function: Pinch stability

Dynamic Function: Thumb extension

Role During Gait: N/A

Role During Throwing: Racquet sports thumb extension

Role During Lifting: Pinch

Synergists: APL, EPL

Antagonists: FPL, flexor pollicis

Length-Tension Relationship: De Quervain component

EMG Data: Moderate pinch EMG

Stretch Position: Flexion

Strength Position: Extension MCP

Common Trigger Points: First compartment

Pain Referral Pattern: Thumb dorsum

Palpation: First compartment

Manual Muscle Test: Thumb extension MMT

Clinical Importance: De Quervain

Common Injuries: Tenosynovitis

Healing Timeline: 4–8 weeks

Contraindications: Spica splint in De Quervain

References: see PubMed [De Quervain EPB]

RECORD: Extensor pollicis longus (EPL)

ID: MUS-UE-021

Name: Extensor pollicis longus (EPL)

Latin Name: Musculus extensor pollicis longus

Region: Forearm

Compartment: Posterior deep
Origin: Ulna interosseous membrane
Insertion: Distal phalanx thumb
Innervation: Posterior interosseous nerve
Nerve Roots: C7–C8
Blood Supply: Posterior interosseous artery
Fascial Connections: Third extensor compartment
Fiber Direction: Oblique
Pennation: Unipennate
Primary Action: Thumb IP extension, MCP extension
Secondary Actions: Wrist extension assist
Stabilizing Function: Retropulsion thumb
Dynamic Function: EPL rupture in RA/hook of hamate fracture
Role During Gait: N/A
Role During Throwing: Thumb extension release
Role During Lifting: Pinch
Synergists: EPB, APL
Antagonists: FPL
Length-Tension Relationship: Crosses Lister tubercle — rupture risk
EMG Data: Moderate
Stretch Position: Flexion
Strength Position: IP extension
Common Trigger Points: Ulnar dorsal forearm
Pain Referral Pattern: Thumb tip
Palpation: EPL tendon ulnar wrist
Manual Muscle Test: IP extension MMT
Clinical Importance: EPL rupture (Mannerfelt)
Common Injuries: Rupture, laceration
Healing Timeline: 8–12 weeks
Contraindications: Protect EPL repair
References: see PubMed [EPL rupture DRUJ]

RECORD: Opponens pollicis

ID: MUS-UE-022
Name: Opponens pollicis
Latin Name: Musculus opponens pollicis
Region: Hand
Compartment: Thenar intrinsic
Origin: Flexor retinaculum, trapezium
Insertion: 1st metacarpal radial shaft
Innervation: Median nerve (recurrent branch)
Nerve Roots: C8–T1
Blood Supply: Superficial palmar arch
Fascial Connections: Thenar fascia

Fiber Direction: Oblique
Pennation: Unipennate
Primary Action: Thumb opposition
Secondary Actions: 1st MC flexion/rotation
Stabilizing Function: CMC stability for pinch
Dynamic Function: Fine motor pinch
Role During Gait: N/A
Role During Throwing: Ball grip opposition
Role During Lifting: Precision grip
Synergists: Abductor pollicis brevis, flexor pollicis brevis
Antagonists: None specific
Length-Tension Relationship: Critical for tip/key pinch
EMG Data: High in opposition
Stretch Position: Extension + abduction
Strength Position: Opposition pad-to-pad
Common Trigger Points: Thenar eminence
Pain Referral Pattern: Thenar/volar thumb
Palpation: Thenar eminence
Manual Muscle Test: Opposition test (pen cap)
Clinical Importance: Median nerve palsy → opposition loss
Common Injuries: Strain rare
Healing Timeline: 6–8 weeks
Contraindications: Protect median motor recovery
References: see PubMed [thenar median nerve]

RECORD: Interossei (Summary)

ID: MUS-UE-023
Name: Interossei (Summary)
Latin Name: Musculi interossei dorsales et palmares
Region: Hand
Compartment: Intrinsic
Origin: Metacarpals
Insertion: Proximal phalanx extensor expansion / base MC
Innervation: Ulnar nerve (palmar/dorsal)
Nerve Roots: C8–T1
Blood Supply: Palmar metacarpal arteries
Fascial Connections: Intermetacarpal fascia
Fiber Direction: Oblique (palmar) / bipennate (dorsal)
Pennation: Mixed
Primary Action: Dorsal: abduction; Palmar: adduction
Secondary Actions: Flexion MCP, extension IP via hood
Stabilizing Function: Metacarpal stability
Dynamic Function: Grip fine tuning
Role During Gait: N/A

Role During Throwing: Finger spread in ball grip
Role During Lifting: Grip strength
Synergists: Lumbricals
Antagonists: None specific
Length-Tension Relationship: Active in grip
EMG Data: Moderate grip EMG
Stretch Position: Full abduction/adduction
Strength Position: Resisted abduction/adduction
Common Trigger Points: Rare
Pain Referral Pattern: Palmar digits
Palpation: Intermetacarpal spaces
Manual Muscle Test: Resisted abduction/adduction
Clinical Importance: Interossei atrophy ulnar nerve sign
Common Injuries: Ulnar neuropathy
Healing Timeline: Variable
Contraindications: None
References: see PubMed [interossei ulnar nerve]

RECORD: Lumbricals (Summary)

ID: MUS-UE-024
Name: Lumbricals (Summary)
Latin Name: Musculi lumbricales I–IV
Region: Hand
Compartment: Intrinsic
Origin: FDP tendons (radial sides)
Insertion: Extensor expansions digits 2–5
Innervation: Median (1–2), Ulnar (3–4)
Nerve Roots: C8–T1
Blood Supply: Palmar digital arteries
Fascial Connections: Palmar fascia
Fiber Direction: Oblique
Pennation: Unipennate
Primary Action: Flexion MCP, extension PIP/DIP
Secondary Actions: Fine motor balance of extensor hood
Stabilizing Function: Synchronized finger curvature
Dynamic Function: Precision grip
Role During Gait: N/A
Role During Throwing: Finger positioning in throw
Role During Lifting: Precision tasks
Synergists: Interossei, FDS
Antagonists: None specific
Length-Tension Relationship: Unique — flex MCP while extend IP
EMG Data: Active fine motor
Stretch Position: MCP extension + IP flexion

Strength Position: MCP flexion IP extension

Common Trigger Points: Palmar hand

Pain Referral Pattern: Palmar digits

Palpation: Observe finger cascade

Manual Muscle Test: Fine motor cascade observation

Clinical Importance: Claw deformity ulnar/median combined

Common Injuries: Rare strain

Healing Timeline: N/A

Contraindications: None

References: see PubMed [lumbrical claw hand]

3. Ligaments

RECORD: Ulnar Collateral Ligament (Elbow) — MCL

ID: LIG-UE-001

Name: Ulnar Collateral Ligament (Elbow) — MCL

Latin Name: Ligamentum collaterale ulnare cubiti

Alternative Names: Medial collateral ligament elbow; UCL elbow

Region: Elbow medial

Joint: Elbow (humeroulnar)

Classification: Extra-articular capsular ligament; anterior/posterior bundles + transverse

Origin: Medial epicondyle humerus

Insertion: Coronoid process ulna (anterior bundle primary)

Fiber Orientation: Anterior bundle resists valgus 20–120° flexion

Primary Restraint: Valgus stress elbow 20–120°

Secondary Restraint: Flexor-pronator mass dynamic stability

Biomechanical Function: Primary static stabilizer against valgus — critical in throwing

Normal Tension: Tightens in valgus and terminal extension

Injury Mechanism: Repetitive valgus overload (throwing) or acute valgus trauma

Associated Injuries: Flexor-pronator tendinopathy, ulnar nerve irritation, avulsion fracture

Clinical Tests: Valgus stress test at 20–30° flexion; moving valgus stress test (throwers)

Grading Systems: Grade I–III sprain; Jobe classification for overhead athletes

Conservative Management: Rest, valgus unloading brace, flexor-pronator strengthening

Surgical Indications: Complete tear in thrower with instability; failed conservative care

Rehabilitation Considerations: Progressive valgus tolerance; avoid early max throwing

Return to Activity Timeline: Throwers 6–12+ months post reconstruction (Tommy John)

Complications: Recurrent instability, flexor contracture, ulnar neuritis

Imaging Recommendations: MRI with valgus stress; ultrasound dynamic

References: Jobe UCL reconstruction; see PubMed [UCL tear throwing athlete]

RECORD: Radial Collateral Ligament (Elbow) — LCL Complex

ID: LIG-UE-002

Name: Radial Collateral Ligament (Elbow) — LCL Complex

Latin Name: Ligamentum collaterale radiale cubiti / lateral ulnar collateral ligament (LUCL)

Alternative Names: Lateral collateral ligament; LUCL key component

Region: Elbow lateral

Joint: Elbow

Classification: Lateral collateral ligamentous complex (RCL, LUCL, annular, accessory)

Origin: Lateral epicondyle

Insertion: Supinator crest ulna (LUCL)

Fiber Orientation: LUCL resists varus and posterolateral rotatory instability

Primary Restraint: Varus stress; posterolateral rotatory stability

Secondary Restraint: Radial head, capsule, extensor origin

Biomechanical Function: Prevents posterolateral subluxation of ulna

Normal Tension: Tight in extension and supination
Injury Mechanism: Fall on outstretched hand with axial load + supination (PLRI)
Associated Injuries: Radial head fracture, elbow dislocation, bone bruise
Clinical Tests: Posterolateral rotatory drawer; varus stress 30° flexion
Grading Systems: Grade I–III; PLRI staging
Conservative Management: Bracing, activity modification if stable
Surgical Indications: Persistent PLRI; LUCL reconstruction
Rehabilitation Considerations: Avoid supinated extension early
Return to Activity Timeline: 3–6 months post reconstruction
Complications: Recurrent PLRI, stiffness
Imaging Recommendations: MRI; varus stress views
References: O'Driscoll PLRI; see PubMed [LUCL posterolateral instability]

RECORD: Annular Ligament

ID: LIG-UE-003
Name: Annular Ligament
Latin Name: Ligamentum anulare radii
Alternative Names: Annular ligament of radius
Region: Elbow proximal
Joint: Proximal radioulnar joint
Classification: Circular band
Origin: Anterior ulnar notch
Insertion: Encircles radial head
Fiber Orientation: Circular
Primary Restraint: Radial head stability in pronation/supination
Secondary Restraint: Quadratus ligament
Biomechanical Function: Maintains radial head contact with ulna during rotation
Normal Tension: Constant around radial neck
Injury Mechanism: Sudden pull on arm (nursemaid's elbow pediatric); dislocation
Associated Injuries: Radial head subluxation in children
Clinical Tests: Supination/pronation assessment; pediatric subluxation history
Grading Systems: N/A — subluxation vs tear
Conservative Management: Reduction pediatric; immobilization if torn
Surgical Indications: Chronic instability rare
Rehabilitation Considerations: Gradual ROM after immobilization
Return to Activity Timeline: Pediatric immediate post reduction; adult 4–6 weeks
Complications: Recurrent subluxation pediatric
Imaging Recommendations: X-ray to exclude fracture; MRI if chronic
References: see PubMed [nursemaid elbow annular ligament]

RECORD: Interosseous Membrane

ID: LIG-UE-004
Name: Interosseous Membrane
Latin Name: Membrana interossea antebrachii

Alternative Names: IOM

Region: Forearm

Joint: Radioulnar syndesmosis

Classification: Fibrous membrane with central band (key structure)

Origin: Radius

Insertion: Ulna

Fiber Orientation: Oblique fibers; central band resists longitudinal displacement

Primary Restraint: Longitudinal stability radius-ulna

Secondary Restraint: DRUJ stability

Biomechanical Function: Force transmission; prevents proximal migration radius

Normal Tension: Tight with axial load

Injury Mechanism: High-energy forearm load; associated with radial head + DRUJ injury (Essex-Lopresti)

Associated Injuries: Radial head fracture, DRUJ disruption, TFCC tear

Clinical Tests: Forearm axial compression pain; DRUJ ballottement

Grading Systems: Essex-Lopresti spectrum

Conservative Management: Rare if isolated mild

Surgical Indications: Essex-Lopresti reconstruction

Rehabilitation Considerations: Long-term forearm strengthening

Return to Activity Timeline: Months if reconstructed

Complications: Proximal radius migration, chronic pain

Imaging Recommendations: MRI forearm; radiographs for radial shortening

References: see PubMed [Essex-Lopresti interosseous membrane]

RECORD: Triangular Fibrocartilage Complex (TFCC)

ID: LIG-UE-005

Name: Triangular Fibrocartilage Complex (TFCC)

Latin Name: Complexus fibrocartilagineus triangularis

Alternative Names: TFCC; ulnar wrist complex

Region: Wrist ulnar

Joint: Distal radioulnar / ulnocarpal

Classification: Complex — TFC disc, meniscus homologue, ulnolunate/ulnotriquetral ligaments, sheath, ECU subsheath

Origin: Ulnar styloid/fovea

Insertion: Lunate/triquetrum/carpals

Fiber Orientation: Central disc avascular; peripheral vascular

Primary Restraint: DRUJ stability; ulnar carpal stability; load transmission

Secondary Restraint: ECU subsheath, interosseous membrane

Biomechanical Function: Distributes ulnar load; allows supination/pronation

Normal Tension: Varies with ulnar variance and grip load

Injury Mechanism: FOOSH, twisting, ulnar positive variance overload

Associated Injuries: DRUJ instability, LT tear, ulnar styloid fracture

Clinical Tests: Fovea sign, TFCC compression test, DRUJ instability

Grading Systems: Palmer classification Class 1 traumatic / Class 2 degenerative

Conservative Management: Immobilization, NSAIDs, strengthening

Surgical Indications: Peripheral tear repair; central debridement; ulnar shortening

Rehabilitation Considerations: Gradual grip/rotation loading
Return to Activity Timeline: 6–12 weeks conservative; 3–6 months post-op
Complications: Chronic ulnar-sided wrist pain, DRUJ arthritis
Imaging Recommendations: MRI arthrogram gold standard; wrist arthroscopy diagnostic
References: Palmer TFCC; see PubMed [TFCC tear treatment]

RECORD: Scapholunate Interosseous Ligament

ID: LIG-UE-006
Name: Scapholunate Interosseous Ligament
Latin Name: Ligamentum scapholunatum
Alternative Names: SL ligament; SLIL
Region: Wrist dorsal central
Joint: Scapholunate
Classification: Intrinsic interosseous — dorsal strongest
Origin: Scaphoid
Insertion: Lunate
Fiber Orientation: Dorsal, proximal, palmar components
Primary Restraint: Scapholunate stability
Secondary Restraint: Extrinsic ligaments (DIC, SLRL)
Biomechanical Function: Maintains scaphoid-lunate alignment; prevents DISI
Normal Tension: Tight in extension and radial deviation
Injury Mechanism: FOOSH hyperextension; high-energy wrist trauma
Associated Injuries: Perilunate dislocation, scaphoid fracture
Clinical Tests: Watson test (scaphoid shift); ballottement; MRI
Grading Systems: Geissler arthroscopic grading I–IV
Conservative Management: Partial tears — immobilization
Surgical Indications: Complete tear with instability; SLAC wrist prevention
Rehabilitation Considerations: Protect wrist extension loading early
Return to Activity Timeline: 3–6 months post repair
Complications: SLAC/SNAC wrist arthritis
Imaging Recommendations: MRI; X-ray Terry Thomas sign, DISI
References: see PubMed [scapholunate ligament tear Watson]

RECORD: Lunotriquetral Interosseous Ligament

ID: LIG-UE-007
Name: Lunotriquetral Interosseous Ligament
Latin Name: Ligamentum lunotriquetrum
Alternative Names: LT ligament
Region: Wrist ulnar proximal row
Joint: Lunotriquetral
Classification: Intrinsic interosseous
Origin: Lunate
Insertion: Triquetrum
Fiber Orientation: Dorsal/palmar/proximal bands

Primary Restraint: LT stability; VISI prevention
Secondary Restraint: Ulnar extrinsic ligaments
Biomechanical Function: Ulnar carpal row stability
Normal Tension: Tight in ulnar deviation
Injury Mechanism: Axial ulnar load, FOOSH
Associated Injuries: TFCC tear, perilunate injury
Clinical Tests: LT ballottement/shuck test; shear test
Grading Systems: Geissler arthroscopic
Conservative Management: Immobilization partial tears
Surgical Indications: Persistent instability, VISI
Rehabilitation Considerations: Similar to SL protocol ulnar side
Return to Activity Timeline: 3–6 months
Complications: Ulnar carpal instability, arthritis
Imaging Recommendations: MRI; VISI on lateral X-ray
References: see PubMed [lunotriquetral ligament tear]

RECORD: Ulnar Collateral Ligament Thumb MCP

ID: LIG-UE-008
Name: Ulnar Collateral Ligament Thumb MCP
Latin Name: Ligamentum collaterale ulnare pollicis
Alternative Names: UCL thumb; gamekeeper's/skier's thumb
Region: Thumb MCP ulnar
Joint: 1st MCP
Classification: Proper and accessory UCL
Origin: Metacarpal head dorsoulnar
Insertion: Proximal phalanx base
Fiber Orientation: Oblique; accessory adds volar stability
Primary Restraint: Valgus stress MCP thumb
Secondary Restraint: Adductor aponeurosis (Stener lesion if interposed)
Biomechanical Function: Pinch stability
Normal Tension: Tight in valgus and flexion
Injury Mechanism: Forced abduction thumb (ski pole, fall)
Associated Injuries: Stener lesion — surgical indication
Clinical Tests: Valgus stress MCP; compare to contralateral
Grading Systems: Grade I–III; Stener present/absent
Conservative Management: Thumb spica if partial, no Stener
Surgical Indications: Complete tear + Stener lesion
Rehabilitation Considerations: Protect valgus 4–6 weeks
Return to Activity Timeline: 6–8 weeks partial; 3 months post-op complete
Complications: Chronic instability, pinch weakness
Imaging Recommendations: Stress X-ray; ultrasound/MRI for Stener
References: see PubMed [UCL thumb Stener lesion]

RECORD: Transverse Carpal Ligament

ID: LIG-UE-009

Name: Transverse Carpal Ligament

Latin Name: Ligamentum carpi transversum / Retinaculum flexorum

Alternative Names: Flexor retinaculum; carpal tunnel roof

Region: Wrist volar

Joint: Carpal tunnel

Classification: Fibrous retinaculum

Origin: Pisiform/hook hamate ulnar

Insertion: Scaphoid/trapezium radial

Fiber Orientation: Transverse

Primary Restraint: Flexor tendons in carpal tunnel

Secondary Restraint: None — compresses contents if thickened

Biomechanical Function: Pulley for flexor tendons; defines carpal tunnel

Normal Tension: Fixed roof — median nerve vulnerable to pressure

Injury Mechanism: Chronic compression (CTS); not typically torn

Associated Injuries: Median neuropathy, flexor tenosynovitis

Clinical Tests: Phalen, Durkan, Tinel carpal tunnel

Grading Systems: CTS severity grading by EMG/ symptoms

Conservative Management: Splint, ergonomics, steroid injection

Surgical Indications: Failed conservative CTS — release retinaculum

Rehabilitation Considerations: Scar management post release; gradual grip

Return to Activity Timeline: 2–6 weeks post carpal tunnel release

Complications: Incomplete release, pillar pain, nerve injury rare

Imaging Recommendations: Ultrasound/MRI if mass lesion; EMG/NCS diagnostic

References: AAOS CTS guideline; see PubMed [carpal tunnel release outcomes]

4. Tendons

RECORD: Extensor Carpi Radialis Brevis (ECRB) Tendon

ID: TEN-UE-001

Name: Extensor Carpi Radialis Brevis (ECRB) Tendon

Latin Name: Tendo musculi extensoris carpi radialis brevis

Alternative Names: ECRB origin tendon

Region: Lateral elbow

Muscle: ECRB

Classification: Common extensor tendon origin

Origin: Lateral epicondyle

Insertion: Base 3rd metacarpal

Synovial Sheath: Origin region prone to angiofibroblastic degeneration

Blood Supply: Hypovascular zone 1–2 cm distal origin

Biomechanical Function: Wrist extension during grip

Pulley System: N/A at origin

Common Pathology: Lateral epicondylalgia (tennis elbow)

Mechanism of Injury: Repetitive wrist extension + grip

Clinical Presentation: Lateral epicondyle pain, grip weakness

Special Tests: Cozen, Maudsley, Mills

Conservative Treatment: Load management, eccentric wrist extension, counterforce brace

Surgical Options: Debridement, ECRB release if refractory

Rehabilitation Protocol: Progressive tendon loading Tyler twist, grip strengthening

Return to Activity: 6–12 weeks to months depending chronicity

Complications: Chronic pain, iatrogenic instability rare

Imaging: Ultrasound hypoechoic origin; MRI

References: Cochrane tennis elbow PMID 23813461 conceptually; see PubMed [lateral epicondylitis]

RECORD: Common Extensor Tendon

ID: TEN-UE-002

Name: Common Extensor Tendon

Latin Name: Tendo extensorum communis originis

Alternative Names: Common extensor origin CEO

Region: Lateral epicondyle

Muscle: ECRB, EDC, EDML, extensor digiti minimi

Classification: Conjoined tendon origin

Origin: Lateral epicondyle

Insertion: Various extensor insertions

Synovial Sheath: No sheath at origin

Blood Supply: Relative hypovascularity at origin

Biomechanical Function: Shared origin for wrist/finger extensors

Pulley System: N/A

Common Pathology: Lateral epicondylalgia

Mechanism of Injury: Overuse gripping sports/tools
Clinical Presentation: Lateral elbow pain radiating forearm
Special Tests: Cozen, grip testing painful
Conservative Treatment: Same as ECRB tendinopathy
Surgical Options: Common extensor release/debridement
Rehabilitation Protocol: Eccentric and isometric loading progression
Return to Activity: Graduated return grip sports
Complications: Prolonged disability 6–24 months if chronic
Imaging: Ultrasound/MRI origin signal change
References: see PubMed [common extensor tendinopathy]

RECORD: Common Flexor Tendon Origin

ID: TEN-UE-003
Name: Common Flexor Tendon Origin
Latin Name: Tendo flexorum communis originis
Alternative Names: Common flexor origin CFO; medial epicondyle tendinopathy
Region: Medial elbow
Muscle: FCR, FCU, FDS, palmaris longus
Classification: Conjoined flexor-pronator origin
Origin: Medial epicondyle
Insertion: Various flexor insertions
Synovial Sheath: N/A at origin
Blood Supply: Hypovascular zone similar to extensors
Biomechanical Function: Wrist/finger flexion and pronation
Pulley System: N/A
Common Pathology: Medial epicondylalgia (golfer's elbow)
Mechanism of Injury: Repetitive flexion/pronation/grip
Clinical Presentation: Medial epicondyle pain, grip pain
Special Tests: Resisted wrist flexion/pronation
Conservative Treatment: Load modification, eccentric flexion, forearm strengthening
Surgical Options: Rare — debridement if refractory
Rehabilitation Protocol: Progressive flexor loading
Return to Activity: 6–12 weeks typical
Complications: Ulnar nerve irritation adjacent
Imaging: Ultrasound/MRI
References: see PubMed [medial epicondylitis golfer elbow]

RECORD: Flexor Pollicis Longus (FPL) Tendon

ID: TEN-UE-004
Name: Flexor Pollicis Longus (FPL) Tendon
Latin Name: Tendo musculi flexoris pollicis longi
Alternative Names: FPL tendon
Region: Forearm/hand thumb
Muscle: Flexor pollicis longus

Classification: Extrinsic flexor — single tendon thumb
Origin: Radius/interosseous membrane
Insertion: Distal phalanx thumb
Synovial Sheath: Thumb flexor sheath
Blood Supply: Vincula and digital arteries
Biomechanical Function: Thumb IP flexion; key pinch power
Pulley System: A1 pulley thumb
Common Pathology: Trigger thumb; laceration
Mechanism of Injury: Overuse; direct laceration zone II
Clinical Presentation: Locking thumb; inability flex IP
Special Tests: FPL isolation (IP flex with MCP blocked)
Conservative Treatment: Splint, corticosteroid injection trigger
Surgical Options: A1 release; tendon repair laceration
Rehabilitation Protocol: Kleinert/Duran if repaired
Return to Activity: 4–8 weeks trigger; 8–12 weeks repair
Complications: Rupture post injection rare; adhesions post repair
Imaging: Ultrasound pulley thickening
References: see PubMed [FPL trigger thumb]

RECORD: Flexor Digitorum Profundus (FDP) — Jersey Finger

ID: TEN-UE-005

Name: Flexor Digitorum Profundus (FDP) — Jersey Finger

Latin Name: Tendo FDP avulsion

Alternative Names: Jersey finger; rugger jersey finger

Region: Ring finger common

Muscle: FDP

Classification: Tendon avulsion from distal phalanx

Origin: Muscle belly forearm

Insertion: Distal phalanx volar

Synovial Sheath: Flexor sheath digit

Blood Supply: Digital arteries via vincula

Biomechanical Function: DIP flexion power grip

Pulley System: A2/A4 critical pulleys

Common Pathology: Avulsion fracture Leddy-Packer classification

Mechanism of Injury: Forced extension active flexion — tackle jersey pull

Clinical Presentation: DIP extension posture; inability isolated DIP flexion

Special Tests: Isolated DIP flexion absent

Conservative Treatment: Zone I partial — extension block splint 6–8 weeks

Surgical Options: Reattachment if retracted (Leddy I/II)

Rehabilitation Protocol: Protected flexion post repair 8–12 weeks

Return to Activity: 3–4 months contact sports

Complications: Fixed flexion deformity, re-rupture

Imaging: Lateral X-ray fleck avulsion; MRI

References: Leddy-Packer classification; see PubMed [jersey finger treatment]

RECORD: Terminal Extensor Tendon — Mallet Finger

ID: TEN-UE-006

Name: Terminal Extensor Tendon — Mallet Finger

Latin Name: Tendo extensorius terminalis

Alternative Names: Extensor tendon insertion DIP; mallet deformity

Region: DIP joint digits

Muscle: Terminal extensor via hood

Classification: Terminal extensor slip

Origin: Lateral bands merge

Insertion: Base distal phalanx dorsal

Synovial Sheath: No sheath

Blood Supply: Periosteal vessels

Biomechanical Function: DIP extension

Pulley System: Extensor mechanism

Common Pathology: Extensor avulsion (tendinous or bony)

Mechanism of Injury: Ball strike fingertip forced flexion

Clinical Presentation: DIP flexed posture; unable actively extend DIP

Special Tests: Elson test — PIP flexion causes rigid DIP in complete tear

Conservative Treatment: Continuous DIP extension splint 6–8 weeks

Surgical Options: Pinning if large bony avulsion or failed conservative

Rehabilitation Protocol: Splint compliance critical; night splint 2–4 more weeks

Return to Activity: 8–10 weeks sport with splint protection

Complications: Swan neck deformity, extensor lag

Imaging: Lateral X-ray bony mallet

References: see PubMed [mallet finger splint Elson]

RECORD: APL/EPB Tendons — De Quervain

ID: TEN-UE-007

Name: APL/EPB Tendons — De Quervain

Latin Name: Tendines compartimenti primi dorsalis

Alternative Names: First dorsal compartment tendons

Region: Radial wrist

Muscle: APL and EPB

Classification: Synovial tendons first extensor compartment

Origin: Forearm muscles

Insertion: Thumb metacarpal/phalanx

Synovial Sheath: First dorsal compartment — stenosing tenosynovitis

Blood Supply: Radial artery branches

Biomechanical Function: Thumb abduction/extension

Pulley System: First compartment retinaculum

Common Pathology: De Quervain tenosynovitis

Mechanism of Injury: Repetitive thumb/wrist ulnar deviation (lifting child, texting)

Clinical Presentation: Radial wrist pain; Finkelstein positive

Special Tests: Finkelstein test

Conservative Treatment: Thumb spica splint, activity modification, injection

Surgical Options: Compartment release if failed conservative

Rehabilitation Protocol: Gradual thumb loading post splint/injection

Return to Activity: 4–6 weeks

Complications: Compartment injury to sensory branch radial nerve

Imaging: Ultrasound thickening/compartment stenosis

References: see PubMed [De Quervain tenosynovitis Finkelstein]

5. Nerves

RECORD: Median Nerve

ID: NRV-UE-001

Name: Median Nerve

Latin Name: Nervus medianus

Alternative Names: N/A

Region: Arm, forearm, hand

Origin: Lateral and medial cords brachial plexus (C5–T1)

Course: Arm lateral to brachial artery; no branches arm; enters forearm via lacertus fibrosus; carpal tunnel

Motor Branches: Pronator teres, FCR, PL, FDS, FPL, APB, OP, lumbricals 1–2

Sensory Distribution: Palmar thumb, index, middle, radial half ring; dorsal fingertips

Cutaneous Landmarks: Thenar eminence (motor); autonomous zone index fingertip pulp

Common Entrapment Sites: Ligament of Struthers, lacertus fibrosus, pronator teres (pronator syndrome), carpal tunnel

Mechanism of Injury: Compression repetitive wrist flexion; trauma laceration wrist

Clinical Deficit Pattern: Thenar weakness, numbness 1–3 digits; positive Phalen/Tinel; ape hand late

Special Tests: Phalen, Durkan, Tinel carpal tunnel, hand elevation test

Neurodynamic Tests: Median nerve bias: wrist extension + elbow extension + shoulder abduction

EMG/NCS Findings: Delayed distal motor latency APB; sensory conduction across carpal tunnel

Differential Diagnosis: C6 radiculopathy, pronator syndrome, cervical stenosis, DM neuropathy

Conservative Management: Night splint neutral wrist, ergonomics, tendon glides, steroid injection

Surgical Indications: Failed conservative, thenar atrophy, persistent EMG changes

Rehabilitation: Nerve glides post release; gradual grip strengthening

Prognosis: Good with early treatment; chronic CTS may have residual numbness

References: AAOS CTS clinical practice guideline; see PubMed [median nerve carpal tunnel]

RECORD: Ulnar Nerve

ID: NRV-UE-002

Name: Ulnar Nerve

Latin Name: Nervus ulnaris

Alternative Names: N/A

Region: Arm, forearm, hand

Origin: Medial cord brachial plexus (C8–T1, T1 primary)

Course: Posterior arm (arcade of Struthers); cubital tunnel posterior medial epicondyle; Guyon's canal wrist

Motor Branches: FCU, FDP 4–5, hypothenar, interossei, adductor pollicis, lumbricals 3–4

Sensory Distribution: Palmar/ dorsal ulnar hand, 5th digit, ulnar half ring

Cutaneous Landmarks: Autonomous zone 5th fingertip pulp

Common Entrapment Sites: Cubital tunnel (most common), Guyon's canal (cyclists/hook hamate)

Mechanism of Injury: Prolonged elbow flexion, leaning on elbow, handlebar compression

Clinical Deficit Pattern: Claw hand, Froment sign, interossei atrophy, ulnar numbness

Special Tests: Tinel elbow, elbow flexion compression test, Froment, Wartenberg

Neurodynamic Tests: Ulnar bias: wrist extension + forearm supination + elbow flexion + shoulder depression

EMG/NCS Findings: Slowing across elbow segment; denervation interossei/ ADM

Differential Diagnosis: C8/T1 radiculopathy, thoracic outlet, Guyon's vs cubital tunnel

Conservative Management: Night elbow extension splint, activity modification, nerve glides

Surgical Indications: Progressive weakness, failed conservative 3–6 months

Rehabilitation: Ulnar nerve glides; grip/pinch retraining

Prognosis: Better if decompressed before severe atrophy

References: see PubMed [cubital tunnel syndrome treatment]

RECORD: Radial Nerve

ID: NRV-UE-003

Name: Radial Nerve

Latin Name: Nervus radialis

Alternative Names: N/A

Region: Arm, forearm, hand (sensory)

Origin: Posterior cord brachial plexus (C5–T8)

Course: Spiral groove humerus; anterior arm; divides PIN and superficial radial at elbow

Motor Branches: Triceps, brachioradialis, ECRL, PIN → extensor forearm/hand

Sensory Distribution: Superficial radial: dorsoradial hand, thumb/index dorsum

Cutaneous Landmarks: First web space dorsal (autonomous zone overlap minimal)

Common Entrapment Sites: Spiral groove (Saturday night palsy), PIN at arcade of Frohse (radial tunnel)

Mechanism of Injury: Compression arm over chair; PIN compression repetitive supination/pronation

Clinical Deficit Pattern: Wrist drop (PIN), triceps spared in spiral groove high lesion

Special Tests: Tinel spiral groove; resisted supination middle finger (radial tunnel)

Neurodynamic Tests: Radial bias: wrist flexion + ulnar deviation + elbow extension + shoulder abduction

EMG/NCS Findings: Denervation extensor muscles; PIN vs SRN localization

Differential Diagnosis: C7 radiculopathy, central lesion, posterior interosseous syndrome

Conservative Management: Splint wrist extension; activity modification; nerve glides

Surgical Indications: Open laceration, failed compression neuropathy decompression

Rehabilitation: Radial nerve glides; wrist/finger extensor strengthening

Prognosis: Saturday night palsy often recovers 6–12 weeks

References: see PubMed [radial nerve palsy spiral groove]

RECORD: Axillary Nerve

ID: NRV-UE-004

Name: Axillary Nerve

Latin Name: Nervus axillaris

Alternative Names: Circumflex nerve

Region: Shoulder

Origin: Posterior cord (C5–C6)

Course: Posterior axillary wall; quadrangular space with posterior circumflex humeral artery

Motor Branches: Deltoid, teres minor

Sensory Distribution: Regimental badge area lateral upper arm

Cutaneous Landmarks: Lateral arm over deltoid insertion

Common Entrapment Sites: Quadrangular space compression; anterior shoulder dislocation

Mechanism of Injury: Shoulder dislocation, humeral neck fracture, compression

Clinical Deficit Pattern: Deltoid weakness, numbness regimental badge, flat shoulder contour if chronic
Special Tests: Abduction MMT; sensation regimental badge
Neurodynamic Tests: Less commonly used; shoulder depression + abduction may tension nerve
EMG/NCS Findings: Denervation deltoid; axillary motor study
Differential Diagnosis: Suprascapular neuropathy, C5 radiculopathy, rotator cuff tear
Conservative Management: Protect shoulder; avoid compression; gradual deltoid strengthening
Surgical Indications: Open injury, quadrangular space syndrome refractory
Rehabilitation: Scapular stabilization; deltoid progressive loading
Prognosis: Often recovers post dislocation unless stretch injury severe
References: see PubMed [axillary nerve injury shoulder dislocation]

RECORD: Musculocutaneous Nerve

ID: NRV-UE-005

Name: Musculocutaneous Nerve

Latin Name: Nervus musculocutaneus

Alternative Names: N/A

Region: Arm, forearm sensory

Origin: Lateral cord (C5–C7)

Course: Through coracobrachialis; lateral arm; becomes lateral antebrachial cutaneous nerve

Motor Branches: Coracobrachialis, biceps, brachialis

Sensory Distribution: Lateral forearm via LACN

Cutaneous Landmarks: Lateral forearm

Common Entrapment Sites: Rare — entrapment in coracobrachialis hypertrophy

Mechanism of Injury: Weightlifting hypertrophy; shoulder surgery iatrogenic

Clinical Deficit Pattern: Elbow flexion weakness; lateral forearm numbness

Special Tests: Elbow flexion/supination MMT; sensation lateral forearm

Neurodynamic Tests: Shoulder extension + elbow extension may tension nerve

EMG/NCS Findings: Denervation biceps/brachialis

Differential Diagnosis: C5/C6 radiculopathy, biceps tendon rupture

Conservative Management: Modify lifting; gradual flexor strengthening

Surgical Indications: Rare decompression

Rehabilitation: Biceps/brachialis strengthening

Prognosis: Generally good with conservative care

References: see PubMed [musculocutaneous nerve entrapment]

6. Joints

RECORD: Glenohumeral Joint

ID: JNT-UE-001

Name: Glenohumeral Joint

Latin Name: Articulatio humeri

Alternative Names: Shoulder joint

Region: Shoulder

Classification: Synovial ball-and-socket

Articular Surfaces: Humeral head, glenoid fossa (labrum deepens)

Joint Capsule: Loose capsule; rotator cuff interval anterior

Primary Ligaments: Superior/middle/inferior GH ligaments

Accessory Ligaments: Coracohumeral, coracoacromial arch

Bursae: Subacromial-subdeltoid bursa

Axis of Motion: Multi-axial

Degrees of Freedom: 3 rotational

Normal ROM: Flexion 180°, ER 90°, IR 70–90°, abduction 180°

Close-Packed Position: Abduction + ER (apprehension position)

Loose-Packed Position: Arm at side slight abduction

Arthrokinematics: Roll-glide varies by quadrant

Osteokinematics: Scapulohumeral rhythm 2:1 elevation

Force Couples: Rotator cuff vs deltoid superior migration control

Common Instability Patterns: Anterior (most common), multidirectional, posterior

Common Stiffness Patterns: Frozen shoulder adhesive capsulitis

Weight-Bearing Role: Limited unless WB upper extremity tasks

Clinical Examination: ROM, impingement signs, instability testing, cuff strength

Imaging: X-ray outlet view; MRI arthrogram instability

References: see PubMed [glenohumeral instability examination]

RECORD: Acromioclavicular (AC) Joint

ID: JNT-UE-002

Name: Acromioclavicular (AC) Joint

Latin Name: Articulatio acromioclavicularis

Alternative Names: See references

Region: Shoulder top

Classification: Synovial plane joint

Articular Surfaces: Acromion, clavicle

Joint Capsule: AC capsule

Primary Ligaments: AC ligaments

Accessory Ligaments: Coracoclavicular (conoid/trapezoid)

Bursae: Minimal

Axis of Motion: Rotation and translation small amplitude

Degrees of Freedom: 3 mm translation, few degrees rotation

Normal ROM: 20° rotation, small AP translation
Close-Packed Position: Arm elevation load
Loose-Packed Position: Arm at rest
Arthrokinematics: Plane gliding
Osteokinematics: Scapular motion with clavicle
Force Couples: Trapezius/deltoid via clavicle
Common Instability Patterns: Rockwood AC separation grades I–VI
Common Stiffness Patterns: Post-traumatic AC arthritis
Weight-Bearing Role: Load transmission clavicle to scapula
Clinical Examination: Cross-body adduction pain, AC tenderness, step-off
Imaging: Zanca AP, weighted views for separation
References: Rockwood AC classification

RECORD: Sternoclavicular (SC) Joint

ID: JNT-UE-003
Name: Sternoclavicular (SC) Joint
Latin Name: Articulatio sternoclavicularis
Alternative Names: See references
Region: Medial shoulder
Classification: Synovial saddle joint
Articular Surfaces: Clavicle, manubrium, 1st costal cartilage
Joint Capsule: See references
Primary Ligaments: SC capsular, interclavicular, costoclavicular
Accessory Ligaments: See references
Bursae: See references
Axis of Motion: See references
Degrees of Freedom: See references
Normal ROM: 35° elevation, 35° AP movement, 50° rotation
Close-Packed Position: Full elevation
Loose-Packed Position: Rest
Arthrokinematics: See references
Osteokinematics: See references
Force Couples: See references
Common Instability Patterns: Posterior dislocation (airway risk), anterior subluxation
Common Stiffness Patterns: See references
Weight-Bearing Role: See references
Clinical Examination: SC prominence, swelling, painful arc
Imaging: Serendipity view; CT for posterior dislocation
References: see PubMed [SC joint dislocation]

RECORD: Elbow — Humeroulnar / Humeroradial / Proximal Radioulnar

ID: JNT-UE-004
Name: Elbow — Humeroulnar / Humeroradial / Proximal Radioulnar
Latin Name: Articulatio cubiti

Alternative Names: See references

Region: Elbow

Classification: Compound joint — hinge + pivot

Articular Surfaces: Trochlea-ulna, capitulum-radius, radial head-ulna

Joint Capsule: See references

Primary Ligaments: UCL, LCL complex, annular ligament

Accessory Ligaments: See references

Bursae: See references

Axis of Motion: See references

Degrees of Freedom: See references

Normal ROM: Flexion 140–150°, extension 0–10° hyperextension; pronation 80°, supination 80°

Close-Packed Position: Full extension supination

Loose-Packed Position: 70° flexion, mid-pronation

Arthrokinematics: See references

Osteokinematics: See references

Force Couples: See references

Common Instability Patterns: Valgus (UCL), PLRI (LUCL)

Common Stiffness Patterns: Post-traumatic contracture

Weight-Bearing Role: See references

Clinical Examination: Valgus/varus stress, pivot shift elbow, ROM

Imaging: X-ray AP/lateral; MRI UCL throwers

References: see PubMed [elbow biomechanics valgus]

RECORD: Distal Radioulnar Joint (DRUJ)

ID: JNT-UE-005

Name: Distal Radioulnar Joint (DRUJ)

Latin Name: Articulatio radioulnaris distalis

Alternative Names: See references

Region: Wrist ulnar

Classification: Pivot joint

Articular Surfaces: See references

Joint Capsule: See references

Primary Ligaments: TFCC, DRUJ capsule

Accessory Ligaments: See references

Bursae: See references

Axis of Motion: See references

Degrees of Freedom: See references

Normal ROM: Supination/pronation distal contribution

Close-Packed Position: Supination

Loose-Packed Position: See references

Arthrokinematics: See references

Osteokinematics: See references

Force Couples: See references

Common Instability Patterns: DRUJ instability post radius fracture

Common Stiffness Patterns: See references
Weight-Bearing Role: See references
Clinical Examination: Piano key test, ballottement ulna, TFCC loading
Imaging: MRI TFCC; comparative DRUJ stress views
References: see PubMed [DRUJ instability]

RECORD: Radiocarpal Joint

ID: JNT-UE-006
Name: Radiocarpal Joint
Latin Name: Articulatio radiocarpalis
Alternative Names: See references
Region: Wrist
Classification: Ellipsoid synovial
Articular Surfaces: Radius scaphoid/lunate, proximal carpal row
Joint Capsule: See references
Primary Ligaments: See references
Accessory Ligaments: See references
Bursae: See references
Axis of Motion: See references
Degrees of Freedom: See references
Normal ROM: Flexion 80°, extension 70°, radial deviation 20°, ulnar deviation 30°
Close-Packed Position: Extension + radial deviation
Loose-Packed Position: Neutral slight flexion
Arthrokinematics: See references
Osteokinematics: See references
Force Couples: See references
Common Instability Patterns: SL/LT dissociation
Common Stiffness Patterns: See references
Weight-Bearing Role: See references
Clinical Examination: Watson test, ROM, grip
Imaging: PA/lateral wrist; MRI ligaments
References: see PubMed [wrist radiocarpal biomechanics]

RECORD: Midcarpal Joint

ID: JNT-UE-007
Name: Midcarpal Joint
Latin Name: Articulatio mediocarpalis
Alternative Names: See references
Region: Wrist
Classification: Complex intercarpal
Articular Surfaces: See references
Joint Capsule: See references
Primary Ligaments: See references
Accessory Ligaments: See references

Bursae: See references
Axis of Motion: See references
Degrees of Freedom: See references
Normal ROM: Contributes ~60% wrist motion
Close-Packed Position: See references
Loose-Packed Position: See references
Arthrokinematics: See references
Osteokinematics: See references
Force Couples: See references
Common Instability Patterns: Perilunate dislocation
Common Stiffness Patterns: See references
Weight-Bearing Role: See references
Clinical Examination: Midcarpal shift test
Imaging: CT/MRI carpal alignment
References: Mayfield perilunate

RECORD: 1st Carpometacarpal (CMC) Joint

ID: JNT-UE-008
Name: 1st Carpometacarpal (CMC) Joint
Latin Name: Articulatio carpometacarpalis pollicis
Alternative Names: See references
Region: Thumb base
Classification: Saddle joint
Articular Surfaces: See references
Joint Capsule: See references
Primary Ligaments: See references
Accessory Ligaments: See references
Bursae: See references
Axis of Motion: See references
Degrees of Freedom: See references
Normal ROM: Opposition, flexion, abduction, adduction
Close-Packed Position: Opposition
Loose-Packed Position: See references
Arthrokinematics: See references
Osteokinematics: See references
Force Couples: See references
Common Instability Patterns: See references
Common Stiffness Patterns: CMC osteoarthritis Eaton stage
Weight-Bearing Role: See references
Clinical Examination: Grind test, opposition strength
Imaging: Robert view
References: Eaton thumb CMC OA

RECORD: Metacarpophalangeal (MCP) Joint

ID: JNT-UE-009

Name: Metacarpophalangeal (MCP) Joint

Latin Name: Articulatio metacarpophalangealis

Alternative Names: See references

Region: Knuckles

Classification: Condylod synovial

Articular Surfaces: See references

Joint Capsule: See references

Primary Ligaments: See references

Accessory Ligaments: See references

Bursae: See references

Axis of Motion: See references

Degrees of Freedom: See references

Normal ROM: Flexion 90°, extension 20–30°, ab/adduction

Close-Packed Position: Full extension

Loose-Packed Position: Flexion

Arthrokinematics: See references

Osteokinematics: See references

Force Couples: See references

Common Instability Patterns: Collateral rupture (thumb UCL)

Common Stiffness Patterns: See references

Weight-Bearing Role: See references

Clinical Examination: Collateral stress

Imaging: See references

References: see PubMed [MCP collateral injury]

RECORD: Proximal Interphalangeal (PIP) Joint

ID: JNT-UE-010

Name: Proximal Interphalangeal (PIP) Joint

Latin Name: Articulatio interphalangealis proximalis

Alternative Names: See references

Region: Finger middle joint

Classification: Hinge synovial

Articular Surfaces: See references

Joint Capsule: See references

Primary Ligaments: See references

Accessory Ligaments: See references

Bursae: See references

Axis of Motion: See references

Degrees of Freedom: See references

Normal ROM: Flexion 100°, extension 0°

Close-Packed Position: See references

Loose-Packed Position: See references

Arthrokinematics: See references

Osteokinematics: See references
Force Couples: See references
Common Instability Patterns: See references
Common Stiffness Patterns: Contracture post injury/surgery
Weight-Bearing Role: See references
Clinical Examination: Bunnell test FDS
Imaging: See references
References: see PubMed [PIP contracture rehabilitation]

RECORD: Distal Interphalangeal (DIP) Joint

ID: JNT-UE-011
Name: Distal Interphalangeal (DIP) Joint
Latin Name: Articulatio interphalangealis distalis
Alternative Names: See references
Region: Finger distal joint
Classification: Hinge synovial
Articular Surfaces: See references
Joint Capsule: See references
Primary Ligaments: See references
Accessory Ligaments: See references
Bursae: See references
Axis of Motion: See references
Degrees of Freedom: See references
Normal ROM: Flexion 80°, extension 0–10°
Close-Packed Position: See references
Loose-Packed Position: See references
Arthrokinematics: See references
Osteokinematics: See references
Force Couples: See references
Common Instability Patterns: Mallet finger extensor disruption
Common Stiffness Patterns: See references
Weight-Bearing Role: See references
Clinical Examination: Elson test
Imaging: See references
References: see PubMed [DIP mallet finger]

7. Biomechanics

RECORD: Shoulder Elevation

ID: BIO-UE-001

Movement Name: Shoulder Elevation

Region: Shoulder

Primary Joints: Glenohumeral, scapulothoracic, AC, SC

Primary Muscles: Deltoid, supraspinatus, trapezius, serratus anterior

Secondary Muscles: Rotator cuff, levator scapulae

Agonists: Deltoid, supraspinatus

Antagonists: Latissimus, pec major

Synergists: Serratus, trapezius for upward rotation

Closed Kinetic Chain Role: Push-up shoulder elevation closed chain

Open Kinetic Chain Role: Overhead reach, lifting

Normal ROM: 180° flexion/abduction

End-Feel: Firm capsular

Common Compensations: Scapular hiking, lumbar extension

Sport-Specific Demands: Swimming, volleyball, tennis serve

Pathological Patterns: Painful arc impingement 60–120°

Assessment Methods: Scapular dyskinesis evaluation, painful arc

Rehabilitation Focus: Scapular control, cuff balance, thoracic mobility

References: see PubMed [scapulohumeral rhythm]

RECORD: Shoulder External/Internal Rotation

ID: BIO-UE-002

Movement Name: Shoulder External/Internal Rotation

Region: Shoulder

Primary Joints: Glenohumeral

Primary Muscles: Infraspinatus, teres minor (ER); subscapularis (IR)

Secondary Muscles: See references

Agonists: See references

Antagonists: See references

Synergists: See references

Closed Kinetic Chain Role: See references

Open Kinetic Chain Role: See references

Normal ROM: ER 90° abducted; IR 70–90°

End-Feel: See references

Common Compensations: See references

Sport-Specific Demands: Throwing, tennis serve

Pathological Patterns: GIRD in throwers; internal impingement

Assessment Methods: See references

Rehabilitation Focus: Posterior capsule mobility, cuff strengthening

References: see PubMed [GIRD throwing shoulder]

RECORD: Elbow Flexion/Extension

ID: BIO-UE-003

Movement Name: Elbow Flexion/Extension

Region: Elbow

Primary Joints: Humeroulnar primary

Primary Muscles: Brachialis, biceps, brachioradialis (flexion); triceps (extension)

Secondary Muscles: See references

Agonists: See references

Antagonists: See references

Synergists: See references

Closed Kinetic Chain Role: See references

Open Kinetic Chain Role: See references

Normal ROM: 0–140° flexion

End-Feel: Soft tissue flexion, bony extension

Common Compensations: See references

Sport-Specific Demands: Throwing, racquet sports, lifting

Pathological Patterns: Flexion contracture post trauma

Assessment Methods: See references

Rehabilitation Focus: Restore extension especially post immobilization

References: see PubMed [elbow ROM rehabilitation]

RECORD: Forearm Pronation/Supination

ID: BIO-UE-004

Movement Name: Forearm Pronation/Supination

Region: Forearm

Primary Joints: PRUJ, DRUJ

Primary Muscles: Pronator teres/quadratus; biceps, supinator

Secondary Muscles: See references

Agonists: See references

Antagonists: See references

Synergists: See references

Closed Kinetic Chain Role: See references

Open Kinetic Chain Role: See references

Normal ROM: 80° each direction

End-Feel: See references

Common Compensations: See references

Sport-Specific Demands: Throwing, racket sports

Pathological Patterns: Essex-Lopresti, annular ligament injury pediatric

Assessment Methods: See references

Rehabilitation Focus: Gradual rotation loading post fracture

References: see PubMed [forearm rotation biomechanics]

RECORD: Wrist Flexion/Extension

ID: BIO-UE-005

Movement Name: Wrist Flexion/Extension

Region: Wrist

Primary Joints: Radiocarpal, midcarpal

Primary Muscles: Flexor carpi group; extensor carpi group

Secondary Muscles: See references

Agonists: See references

Antagonists: See references

Synergists: See references

Closed Kinetic Chain Role: See references

Open Kinetic Chain Role: See references

Normal ROM: Flexion 80°, extension 70°

End-Feel: See references

Common Compensations: See references

Sport-Specific Demands: Golf, tennis, gymnastics

Pathological Patterns: Extensor tendinopathy with repetitive grip extension

Assessment Methods: See references

Rehabilitation Focus: Wrist neutral strengthening, eccentric loading

References: see PubMed [wrist tendinopathy loading]

RECORD: Grip Strength

ID: BIO-UE-006

Movement Name: Grip Strength

Region: Hand

Primary Joints: See references

Primary Muscles: FDS, FDP, lumbricals, interossei, thenar/hypothenar

Secondary Muscles: See references

Agonists: See references

Antagonists: See references

Synergists: See references

Closed Kinetic Chain Role: See references

Open Kinetic Chain Role: See references

Normal ROM: N/A — force measure

End-Feel: See references

Common Compensations: See references

Sport-Specific Demands: Climbing, wrestling, racquet sports

Pathological Patterns: Weakness CTS, ulnar neuropathy, jersey finger

Assessment Methods: Dynamometer Jamar, pinch gauge

Rehabilitation Focus: Graded grip loading, putty exercises

References: see PubMed [grip strength normative data]

RECORD: Overhead Throwing

ID: BIO-UE-007

Movement Name: Overhead Throwing

Region: Shoulder/elbow

Primary Joints: GH, elbow, wrist

Primary Muscles: Kinetic chain — lower extremity through cuff, deltoid, triceps, forearm

Secondary Muscles: See references

Agonists: See references

Antagonists: See references

Synergists: See references

Closed Kinetic Chain Role: See references

Open Kinetic Chain Role: See references

Normal ROM: See references

End-Feel: See references

Common Compensations: See references

Sport-Specific Demands: Baseball, cricket, javelin

Pathological Patterns: UCL injury, internal impingement, labral tears

Assessment Methods: Throwing analysis, shoulder/elbow ROM GIRD

Rehabilitation Focus: Return to throw programs, valgus tolerance

References: ASMI throwing biomechanics; see PubMed [overhead throwing phases]

8. Clinical Examination Template

RECORD: Comprehensive Upper Extremity Clinical Examination

ID: EXAM-UE-001

Template Name: Comprehensive Upper Extremity Clinical Examination

Chief Complaint: Patient-reported primary symptom in own words (e.g., lateral elbow pain with gripping)

History of Present Illness: Onset (acute/chronic), mechanism, location, quality, severity (NRS 0–10), timing, aggravating/alleviating factors, prior treatment, functional limitations (work/sport/ADL)

Past Medical History: Prior UE injury/surgery, diabetes (CTS/neuropathy), inflammatory arthritis, bleeding disorders

Medications: Anticoagulants, steroids, neuropathic agents

Social/Occupational History: Dominant hand, job demands (keyboard, tools, overhead), sport/recreation, smoking

Review of Systems: Cervical symptoms (radiculopathy screen), systemic inflammatory signs, neurologic symptoms distal

Observation: Posture, scapular position, muscle atrophy (thenar/hypothenar/interossei), swelling, deformity, surgical scars

Palpation: Systematic joint lines, epicondyles, snuffbox, carpal tunnel, cubital tunnel, trigger points

Range of Motion: Cervical screen; shoulder active/passive; elbow; forearm rotation; wrist; digits — compare bilaterally

Strength Testing: Manual muscle testing rotator cuff, elbow flexors/extensors, wrist, grip/pinch — note pain vs true weakness

Neurovascular Exam: Median/ulnar/radial sensation, Tinel/Phalen as indicated, pulses, capillary refill digits

Special Tests: Condition-specific: impingement (Neer/Hawkins), instability (apprehension), Cozen/Mills elbow, TFCC, CTS tests

Functional Assessment: Lift, reach overhead, grip task, sport-specific simulation if appropriate

Outcome Measures: DASH, QuickDASH, SPADI, PRTEE (elbow), pain NRS, grip dynamometry

Clinical Impression: Working musculoskeletal diagnosis with tissue/joint identification

Differential Diagnosis: Ranked list including cervical referral, joint vs tendon vs nerve

Working Diagnosis: Primary diagnosis with confidence level; red flags ruled out

Plan: Imaging if indicated, PT referral, activity modification, follow-up timeline

References: DASH outcome measure; APTA examination guidelines; see PubMed [upper extremity clinical examination]

9. Special Tests

RECORD: Cozen Test

ID: TEST-UE-001

Name: Cozen Test

Also Known As: Resisted wrist extension test

Purpose: Screen for lateral epicondylalgia (tennis elbow)

Structures Tested: ECRB/common extensor origin at lateral epicondyle

Technique: Patient seated; elbow extended, forearm pronated, wrist radially deviated; patient makes fist and extends wrist while examiner resists

Positive Finding: Pain over lateral epicondyle

Sensitivity: ~0.84 (varies by study)

Specificity: ~0.29 (low — clinical diagnosis composite)

Clinical Utility: Useful provocative test; combine with palpation and history

Evidence Level: Level C — expert opinion, diagnostic accuracy moderate

References: see PubMed [Cozen test lateral epicondylitis diagnostic accuracy]

RECORD: Mills Test

ID: TEST-UE-002

Name: Mills Test

Purpose: Lateral epicondylalgia

Structures Tested: Common extensor tendon, ECRB

Technique: Examiner passively pronates forearm, flexes wrist and elbow while palpating lateral epicondyle

Positive Finding: Sudden pain at lateral epicondyle

Sensitivity: ~0.53

Specificity: ~0.76

Clinical Utility: Passive alternative when resisted testing difficult

Evidence Level: Level C

References: see PubMed [Mills test tennis elbow]

RECORD: Maudsley Test

ID: TEST-UE-003

Name: Maudsley Test (Middle Finger Extension Test)

Purpose: Lateral epicondylalgia

Structures Tested: ECRB (middle finger extensor connection)

Technique: Elbow extended, forearm pronated; resist extension of middle finger at MCP

Positive Finding: Pain at lateral epicondyle

Sensitivity: ~0.68

Specificity: ~0.41

Clinical Utility: Quick office test; part of cluster

Evidence Level: Level C

References: see PubMed [Maudsley test ECRB]

RECORD: Tinel Sign — Wrist (Carpal Tunnel)

ID: TEST-UE-004

Name: Tinel Sign at Wrist

Purpose: Median nerve irritation at carpal tunnel

Structures Tested: Median nerve beneath transverse carpal ligament

Technique: Percuss over carpal tunnel at wrist crease

Positive Finding: Tingling/paresthesia in median nerve distribution

Sensitivity: ~0.48

Specificity: ~0.77

Clinical Utility: Support CTS diagnosis with Phalen and symptoms

Evidence Level: Level B

References: AAOS CTS guideline

RECORD: Tinel Sign — Elbow (Cubital Tunnel)

ID: TEST-UE-005

Name: Tinel Sign at Elbow

Purpose: Ulnar nerve irritation cubital tunnel

Structures Tested: Ulnar nerve posterior to medial epicondyle

Technique: Percuss ulnar nerve in cubital tunnel

Positive Finding: Paresthesia ulnar digits

Sensitivity: ~0.70

Specificity: ~0.95

Clinical Utility: High specificity supports cubital tunnel

Evidence Level: Level B

References: see PubMed [cubital tunnel Tinel]

RECORD: Phalen Test

ID: TEST-UE-006

Name: Phalen Maneuver

Purpose: Carpal tunnel syndrome screening

Structures Tested: Median nerve compression in carpal tunnel

Technique: Wrists fully flexed 60 seconds (backs of hands together)

Positive Finding: Median nerve paresthesia within 60 sec

Sensitivity: ~0.75

Specificity: ~0.47

Clinical Utility: Classic CTS test; combine with Durkan and EMG

Evidence Level: Level B

References: AAOS CTS; PMID literature Phalen

RECORD: Durkan Compression Test

ID: TEST-UE-007

Name: Durkan Carpal Compression Test

Purpose: CTS screening

Structures Tested: Median nerve in carpal tunnel

Technique: Examiner applies direct pressure over carpal tunnel 30 seconds

Positive Finding: Median paresthesia or pain
Sensitivity: ~0.87
Specificity: ~0.22
Clinical Utility: Sensitive provocative test
Evidence Level: Level C
References: see PubMed [Durkan compression test CTS]

RECORD: Watson Test (Scaphoid Shift)

ID: TEST-UE-008
Name: Watson Test
Purpose: Scapholunate instability
Structures Tested: Scapholunate ligament, scaphoid alignment
Technique: Examiner thumb pressure on scaphoid volarly; wrist ulnar to radial deviation
Positive Finding: Pain, clunk, scaphoid subluxation reducing
Sensitivity: ~0.69
Specificity: ~0.66
Clinical Utility: Important SL tear screen; compare contralateral
Evidence Level: Level B
References: see PubMed [Watson scaphoid shift test]

RECORD: Elson Test

ID: TEST-UE-009
Name: Elson Test
Purpose: Central slip/extensor mechanism integrity at PIP (mallet differential)
Structures Tested: Central slip, lateral bands, terminal extensor
Technique: PIP flexed 90° over edge; extend middle phalanx against resistance; observe DIP
Positive Finding: Complete tear: rigid extended DIP; partial: weak/floppy DIP extension
Sensitivity: High for complete mallet
Specificity: Moderate
Clinical Utility: Differentiate mallet vs other extensor lag
Evidence Level: Level C
References: see PubMed [Elson test mallet finger]

RECORD: Froment Sign

ID: TEST-UE-010
Name: Froment Sign
Purpose: Ulnar nerve motor weakness (adductor pollicis)
Structures Tested: Adductor pollicis, ulnar nerve motor branch
Technique: Patient holds paper between thumb and index with IP flexion; examiner pulls paper
Positive Finding: Thumb IP flexes (FPL substitution) to maintain grip
Sensitivity: ~0.95 for ulnar motor deficit
Specificity: ~0.96
Clinical Utility: Specific for ulnar motor neuropathy
Evidence Level: Level B

References: see PubMed [Froment sign ulnar nerve]

RECORD: Bunnell Test

ID: TEST-UE-011

Name: Bunnell Test

Purpose: FDS integrity per digit

Structures Tested: Flexor digitorum superficialis tendon

Technique: Hold all FDP out except test digit (hold other digits extended blocks FDP); flex test digit PIP

Positive Finding: Unable to flex PIP suggests FDS rupture/laceration

Clinical Utility: Standard flexor tendon assessment

Evidence Level: Level C

References: see PubMed [Bunnell FDS test]

RECORD: Valgus Stress Test (Elbow)

ID: TEST-UE-012

Name: Valgus Stress Test Elbow

Purpose: UCL/MCL elbow integrity

Structures Tested: Ulnar collateral ligament anterior bundle

Technique: Elbow 20–30° flexion (unlocks olecranon); valgus force while supinating forearm

Positive Finding: Pain medial elbow and/or increased laxity vs contralateral

Sensitivity: ~0.78 moving valgus in throwers

Specificity: Variable

Clinical Utility: Essential thrower exam; moving valgus stress test during simulated throw

Evidence Level: Level B

References: see PubMed [UCL valgus stress test throwing]

RECORD: Varus Stress Test (Elbow)

ID: TEST-UE-013

Name: Varus Stress Test Elbow

Purpose: LCL/LUCL integrity, posterolateral rotatory instability

Structures Tested: Lateral collateral ligament complex, LUCL

Technique: Elbow 20–30° flexion; varus force applied

Positive Finding: Lateral pain, increased laxity, apprehension

Clinical Utility: Assess lateral elbow instability post dislocation

Evidence Level: Level C

References: O'Driscoll PLRI; see PubMed [varus stress elbow LUCL]

10. Pathologies

RECORD: Lateral Epicondylalgia (Tennis Elbow)

ID: PATH-UE-001

Name: Lateral Epicondylalgia (Tennis Elbow)

Alternative Names: Tennis elbow; lateral epicondylitis (misnomer — tendinopathy)

ICD-10 Code (approx): M77.1

Region: Lateral elbow

Definition: Degenerative tendinopathy of common extensor origin, primarily ECRB

Epidemiology: 1–3% population; peak 35–55 years; racquet sports, manual labor

Etiology: Repetitive wrist extension and grip; overload microtrauma

Pathophysiology: Angiofibroblastic hyperplasia, not classic inflammation

Risk Factors: Repetitive tool use, poor technique, smoking, age

Typical Presentation: Lateral elbow pain weeks–months; worse gripping, lifting

Clinical Findings: Lateral epicondyle tenderness, pain resisted wrist extension

Special Tests: Cozen, Maudsley, Mills positive

Imaging Findings: US/MRI tendon thickening, hypoechoic origin; X-ray usually normal

Classification/Grading: Acute vs chronic >3 months; severity by function/pain

Differential Diagnosis: Radial tunnel, C6 radiculopathy, lateral ligament sprain, OA

Conservative Management: Activity modification, counterforce brace, wait-and-see vs PT

Medical Management: NSAIDs short course; corticosteroid injection (short-term relief, worse long-term vs PT)

Surgical Indications: 6–12 months failed exhaustive conservative

Physical Therapy Interventions: Eccentric wrist extension (Tyler twist), isometric loading, manual therapy

Exercise Progression: Isometric → eccentric → concentric → sport-specific grip

Return to Sport Criteria: Pain-free grip strength >85% contralateral, sport simulation without pain

Prognosis: 90% improve 6–12 months conservative; recalcitrant cases prolonged

Complications: Chronic pain, iatrogenic tendon rupture post injection rare

Evidence Summary: Cochrane: PT and eccentric loading beneficial; CSI short-term only (Level A)

References: Cochrane tennis elbow PMID 23813461; see PubMed [lateral epicondylalgia rehabilitation]

RECORD: Medial Epicondylalgia (Golfer's Elbow)

ID: PATH-UE-002

Name: Medial Epicondylalgia (Golfer's Elbow)

Alternative Names: Golfer's elbow; medial epicondylitis

ICD-10 Code (approx): M77.0

Region: Medial elbow

Definition: Tendinopathy common flexor-pronator origin

Epidemiology: Less common than lateral; golf, throwing, climbing

Etiology: Repetitive wrist flexion/pronation/grip

Pathophysiology: Similar degenerative tendinopathy ECRB counterpart

Risk Factors: See references

Typical Presentation: Medial elbow pain gripping, throwing, lifting

Clinical Findings: Medial epicondyle tenderness, pain resisted flexion/pronation

Special Tests: Resisted wrist flexion/pronation

Imaging Findings: See references
Classification/Grading: See references
Differential Diagnosis: UCL sprain, cubital tunnel, cervical radiculopathy
Conservative Management: Load management, flexor eccentric exercises, bracing
Medical Management: See references
Surgical Indications: See references
Physical Therapy Interventions: Eccentric flexor/pronator loading, nerve glides if ulnar symptoms
Exercise Progression: See references
Return to Sport Criteria: Pain-free valgus tolerance throwers; grip symmetry
Prognosis: See references
Complications: See references
Evidence Summary: Less RCT data than lateral; eccentric loading extrapolated (Level B)
References: see PubMed [medial epicondylalgia treatment]

RECORD: Cubital Tunnel Syndrome

ID: PATH-UE-003
Name: Cubital Tunnel Syndrome
Alternative Names: Ulnar neuropathy at elbow
ICD-10 Code (approx): G56.2
Region: Elbow/hand
Definition: Compression ulnar nerve cubital tunnel
Epidemiology: Second most common UE entrapment after CTS
Etiology: Prolonged flexion, compression, anatomical variants (arcade Struthers)
Pathophysiology: Ischemic and demyelinating injury ulnar nerve
Risk Factors: See references
Typical Presentation: Ring/small finger numbness, hand weakness, elbow pain
Clinical Findings: Tinel elbow, Froment positive, interossei atrophy late
Special Tests: Elbow flexion compression test, Tinel, Froment
Imaging Findings: EMG/NCS gold standard; MRI if mass lesion
Classification/Grading: See references
Differential Diagnosis: See references
Conservative Management: Night extension splint, activity modification, nerve glides
Medical Management: See references
Surgical Indications: Progressive weakness, EMG axonal changes
Physical Therapy Interventions: Ulnar nerve glides, ergonomic education, strengthening interossei
Exercise Progression: See references
Return to Sport Criteria: Symptom-free sport; strength >90% pinch
Prognosis: See references
Complications: See references
Evidence Summary: Splinting and surgery effective; PT glides adjunct (Level B)
References: see PubMed [cubital tunnel syndrome AAOS]

RECORD: Carpal Tunnel Syndrome

ID: PATH-UE-004

Name: Carpal Tunnel Syndrome

Alternative Names: CTS

ICD-10 Code (approx): G56.0

Region: Wrist/hand

Definition: Median nerve compression within carpal tunnel

Epidemiology: 3–5% adults; female predominance; pregnancy, diabetes, repetitive hand use

Etiology: Tunnel pressure increase — multifactorial

Pathophysiology: Median nerve ischemia/demyelination

Risk Factors: See references

Typical Presentation: Night numbness 1–3 digits, thenar weakness, dropping objects

Clinical Findings: Phalen, Durkan, Tinel positive; thenar atrophy late

Special Tests: Phalen, Durkan, Tinel, hand elevation test

Imaging Findings: EMG delayed distal latency; US enlarged nerve cross-section

Classification/Grading: See references

Differential Diagnosis: See references

Conservative Management: Night splint, ergonomics, tendon/nerve glides, steroid injection

Medical Management: See references

Surgical Indications: Failed conservative, thenar weakness, persistent EMG

Physical Therapy Interventions: Median nerve glides, tendon glides, strengthening post release

Exercise Progression: See references

Return to Sport Criteria: Pain-free grip; symptom resolution

Prognosis: See references

Complications: See references

Evidence Summary: AAOS guideline: local steroid and surgery effective (Level A)

References: AAOS CTS CPG; see PubMed [carpal tunnel splinting]

RECORD: De Quervain Tenosynovitis

ID: PATH-UE-005

Name: De Quervain Tenosynovitis

Alternative Names: First dorsal compartment tenosynovitis

ICD-10 Code (approx): M65.4

Region: Radial wrist

Definition: Stenosing tenosynovitis APL/EPB tendons

Epidemiology: Common new mothers, mobile device use, racquet sports

Etiology: Repetitive thumb/wrist ulnar deviation

Pathophysiology: See references

Risk Factors: See references

Typical Presentation: Radial wrist pain, thumb motion painful

Clinical Findings: Finkelstein test positive, first compartment tenderness

Special Tests: See references

Imaging Findings: See references

Classification/Grading: See references

Differential Diagnosis: See references

Conservative Management: Thumb spica splint, NSAIDs, steroid injection (70–80% success)

Medical Management: See references

Surgical Indications: See references

Physical Therapy Interventions: Activity modification, gradual thumb loading, manual therapy

Exercise Progression: See references

Return to Sport Criteria: Finkelstein negative; pinch without pain

Prognosis: See references

Complications: See references

Evidence Summary: Splint + injection first line (Level B)

References: see PubMed [De Quervain tenosynovitis treatment]

RECORD: Trigger Finger (Stenosing Tenosynovitis)

ID: PATH-UE-006

Name: Trigger Finger (Stenosing Tenosynovitis)

Alternative Names: Trigger digit

ICD-10 Code (approx): M65.3

Region: Hand

Definition: A1 pulley thickening restricting flexor tendon glide

Epidemiology: Middle age, diabetes, repetitive gripping

Etiology: See references

Pathophysiology: See references

Risk Factors: See references

Typical Presentation: Locking, clicking, morning stiffness digit

Clinical Findings: Palpable nodule A1 pulley, triggering extension

Special Tests: See references

Imaging Findings: See references

Classification/Grading: See references

Differential Diagnosis: See references

Conservative Management: Splinting, injection, activity modification

Medical Management: See references

Surgical Indications: Failed 2 injections, fixed contracture

Physical Therapy Interventions: Tendon glides, gentle ROM post injection/release

Exercise Progression: See references

Return to Sport Criteria: Full painless digit flexion/extension

Prognosis: See references

Complications: See references

Evidence Summary: See references

References: see PubMed [trigger finger corticosteroid injection]

RECORD: Mallet Finger

ID: PATH-UE-007

Name: Mallet Finger

Alternative Names: Baseball finger

ICD-10 Code (approx): M20.0 / S66.1

Region: DIP joint

Definition: Terminal extensor tendon avulsion (open or closed)
Epidemiology: Ball sports, manual labor
Etiology: See references
Pathophysiology: See references
Risk Factors: See references
Typical Presentation: DIP flexion deformity, inability extend DIP
Clinical Findings: Elson test positive complete tear; bony avulsion on X-ray
Special Tests: See references
Imaging Findings: See references
Classification/Grading: See references
Differential Diagnosis: See references
Conservative Management: Continuous DIP extension splint 6–8 weeks (even bony if no joint sublux)
Medical Management: See references
Surgical Indications: Large fragment >30% articular, subluxation, open injury
Physical Therapy Interventions: Splint education, skin care, gradual weaning
Exercise Progression: See references
Return to Sport Criteria: 8–10 weeks with splint protection sport
Prognosis: See references
Complications: See references
Evidence Summary: See references
References: see PubMed [mallet finger conservative splint]

RECORD: Jersey Finger

ID: PATH-UE-008
Name: Jersey Finger
Alternative Names: Rugger jersey finger
ICD-10 Code (approx): S66.4
Region: Ring finger common
Definition: FDP avulsion from distal phalanx
Epidemiology: Contact sports — rugby, football
Etiology: See references
Pathophysiology: See references
Risk Factors: See references
Typical Presentation: DIP held extended, unable isolated DIP flexion
Clinical Findings: See references
Special Tests: See references
Imaging Findings: See references
Classification/Grading: Leddy-Packer I–IV by retraction and bony involvement
Differential Diagnosis: See references
Conservative Management: Leddy I partial — extension block splint 6–8 weeks
Medical Management: See references
Surgical Indications: Leddy II–IV retracted — urgent repair
Physical Therapy Interventions: Protected flexion protocol post repair
Exercise Progression: See references

Return to Sport Criteria: 3–4 months contact sport post surgical repair

Prognosis: See references

Complications: See references

Evidence Summary: See references

References: Leddy-Packer; see PubMed [jersey finger surgical outcomes]

RECORD: TFCC Injury

ID: PATH-UE-009

Name: TFCC Injury

Alternative Names: Triangular fibrocartilage complex tear

ICD-10 Code (approx): S63.5

Region: Ulnar wrist

Definition: Traumatic or degenerative TFCC tear

Epidemiology: FOOSH, twisting injury; ulnar positive variance

Etiology: See references

Pathophysiology: See references

Risk Factors: See references

Typical Presentation: Ulnar wrist pain, clicking, weakness grip

Clinical Findings: Fovea sign, TFCC compression pain, DRUJ instability

Special Tests: See references

Imaging Findings: MRI arthrogram sensitive; arthroscopy gold standard

Classification/Grading: Palmer Class 1 traumatic / Class 2 degenerative

Differential Diagnosis: See references

Conservative Management: Immobilization, PT, steroid injection

Medical Management: See references

Surgical Indications: Peripheral tear repair; central debridement; ulnar shortening

Physical Therapy Interventions: Wrist stabilization, gradual rotation/grip loading

Exercise Progression: See references

Return to Sport Criteria: Painless weight-bearing wrist; sport simulation

Prognosis: See references

Complications: See references

Evidence Summary: See references

References: Palmer TFCC; see PubMed [TFCC tear treatment outcomes]

RECORD: Scapholunate Ligament Tear

ID: PATH-UE-010

Name: Scapholunate Ligament Tear

Alternative Names: SL tear; SLIL injury

ICD-10 Code (approx): S63.3

Region: Wrist

Definition: Tear scapholunate interosseous ligament

Epidemiology: FOOSH hyperextension; athletes, falls

Etiology: See references

Pathophysiology: See references

Risk Factors: See references

Typical Presentation: Dorsal wrist pain, weakness, clicking

Clinical Findings: Watson test positive, SL gap, tenderness dorsal

Special Tests: See references

Imaging Findings: Terry Thomas sign, DISI on lateral; MRI/arthroscopy

Classification/Grading: Geissler arthroscopic I–IV

Differential Diagnosis: See references

Conservative Management: Partial tears — immobilization, PT

Medical Management: See references

Surgical Indications: Complete tear with instability — repair/ligament reconstruction

Physical Therapy Interventions: Wrist proprioception, scaphoid stabilization exercises

Exercise Progression: See references

Return to Sport Criteria: 3–6 months post repair; painless loading

Prognosis: See references

Complications: SLAC wrist arthritis if untreated

Evidence Summary: See references

References: see PubMed [scapholunate ligament tear treatment]

11. Rehabilitation Phases

RECORD: Acute / Protective Phase

ID: REHAB-UE-P1

Phase Name: Acute / Protective Phase

Timeframe: 0–2 weeks (injury dependent)

Goals: Protect healing tissue, reduce pain/inflammation, maintain safe ROM

Precautions: Avoid painful arcs, no aggressive stretching acutely tendon/ ligament

Pain Management: Relative rest, ice, NSAIDs per MD, modalities as indicated

ROM: Gentle passive/active-assisted pain-free ranges

Strength: Isometric submaximal pain-free; scapular setting

Neuromuscular: Postural awareness, diaphragmatic breathing

Manual Therapy: Gentle soft tissue, avoid aggressive joint mobs acute

Functional Training: ADL modifications, ergonomic setup

Home Exercise Program: Pendulum, AAROM, isometrics, nerve glides if nerve irritation

Progression Criteria: Decreasing pain, tolerating gentle ROM

References: APTA acute care guidelines; see PubMed [tendinopathy acute phase load]

RECORD: Early Recovery Phase

ID: REHAB-UE-P2

Phase Name: Early Recovery Phase

Timeframe: 2–6 weeks

Goals: Restore full pain-free ROM, begin loading, neuromuscular control

ROM: Active ROM all planes; joint mobs if stiff

Strength: Isotonic light resistance bands; begin eccentric tendinopathy protocols

Neuromuscular: Scapular stabilization, rotator cuff activation

Manual Therapy: Soft tissue, grade I–III mobs

Functional Training: Light functional reaching, grip progression

Home Exercise Program: Band ER/IR, wrist eccentric, nerve glides

Progression Criteria: Full ROM, strength 50%+ contralateral

References: see PubMed [early rehab upper extremity]

RECORD: Intermediate Strength Phase

ID: REHAB-UE-P3

Phase Name: Intermediate Strength Phase

Timeframe: 6–12 weeks

Goals: Strength endurance 70–85% contralateral, sport/work prep

Strength: Progressive resistance, compound UE exercises, plyometric prep

Neuromuscular: Proprioception, closed chain wall push-ups

Functional Training: Simulated work tasks, light sport drills

Progression Criteria: Strength >70%, special tests negative

References: see PubMed [intermediate UE strengthening]

RECORD: Advanced Functional Phase

ID: REHAB-UE-P4

Phase Name: Advanced Functional Phase

Timeframe: 12–16+ weeks

Goals: Sport/work specific power, agility, endurance

Strength: High-load functional patterns, eccentric overload

Functional Training: Sport-specific drills non-contact

Progression Criteria: >85–90% strength, functional tests pass

References: Thrower's ten program; see PubMed [advanced UE rehab]

RECORD: Return to Sport Phase

ID: REHAB-UE-P5

Phase Name: Return to Sport Phase

Timeframe: Variable

Goals: Full RTS/RTW without symptom recurrence

Functional Training: Graduated return protocols (throwing, tennis, golf)

Return Criteria: Objective strength/ROM, sport simulation pain-free, psychological readiness

References: ASMI interval throwing; see PubMed [return to sport upper extremity]

12. Exercise Library

RECORD: Pendulum Exercise (Codman)

ID: EX-UE-001

Name: Pendulum Exercise

Region: Shoulder

Phase: Acute/early recovery

Purpose: Maintain mobility with gravity assistance

Technique: Lean forward, relaxed arm circles, forward/back, side-to-side

Dosage: 2–3 sets 30–60 sec each direction, 2–3x daily

Progression: Increase circle size as tolerated

Precautions: No active shoulder shrug

Evidence: Standard post-op shoulder protocol Level C

References: see PubMed [Codman pendulum shoulder]

RECORD: Isometric External Rotation

ID: EX-UE-002

Name: Isometric External Rotation at Side

Region: Shoulder

Phase: Acute

Purpose: Rotator cuff activation without motion

Technique: Towel roll at elbow, press outward into wall/hand hold 5–10 sec

Dosage: 10 reps x 5–10 sec, 2–3x daily

Precautions: Pain-free submaximal

References: see PubMed [isometric cuff activation]

RECORD: Band External Rotation

ID: EX-UE-003

Name: Resisted External Rotation with Band

Region: Shoulder

Phase: Early–intermediate

Purpose: Strengthen infraspinatus/teres minor

Technique: Elbow 90° at side, rotate arm outward against band

Dosage: 2–3 sets 12–15 reps

Progression: Increase resistance, side-lying ER

References: see PubMed [band ER rotator cuff]

RECORD: Scapular Setting / Retraction

ID: EX-UE-004

Name: Scapular Retraction (Setting)

Region: Scapulothoracic

Phase: All phases

Purpose: Improve scapular control

Technique: Gently squeeze scapulae together and down, hold 5 sec

Dosage: 10–15 reps

References: Kibler scapular protocols

RECORD: Eccentric Wrist Extension (Tyler Twist / Cozen Protocol)

ID: EX-UE-005

Name: Eccentric Wrist Extension (FlexBar Tyler Twist)

Region: Elbow/wrist

Phase: Early recovery lateral epicondylalgia

Purpose: Load ECRB tendon eccentrically

Technique: FlexBar: wrist extension affected side, slow eccentric flexion controlled

Dosage: 3 sets 15 daily

Evidence: Level B — RCT Tyler twist tennis elbow

References: see PubMed [Tyler twist FlexBar tennis elbow]

RECORD: Median Nerve Glide

ID: EX-UE-006

Name: Median Nerve Glide

Region: Arm/hand

Phase: CTS, pronator syndrome

Purpose: Improve nerve mobility

Technique: Wrist extension, elbow extension, shoulder abduction — slider technique

Dosage: 10 slow reps 2–3x daily

Precautions: Symptom monitoring — mild tingling ok, no increase symptoms after

References: see PubMed [median nerve gliding CTS]

RECORD: Ulnar Nerve Glide

ID: EX-UE-007

Name: Ulnar Nerve Glide

Region: Elbow/hand

Phase: Cubital tunnel

Purpose: Ulnar nerve mobility

Technique: Shoulder abduction, elbow flexion, wrist extension sequence

Dosage: 10 reps 2–3x daily

References: see PubMed [ulnar nerve gliding exercises]

RECORD: Grip Strengthening with Putty

ID: EX-UE-008

Name: Therapeutic Putty Grip Exercises

Region: Hand

Phase: Intermediate

Purpose: Grip endurance and strength

Technique: Squeeze, spread, pinch with graded putty

Dosage: 2–3 sets 10–15 reps

Progression: Firmer putty

References: see PubMed [hand putty grip strengthening]

RECORD: Dart Thrower's Motion (DTM)

ID: EX-UE-009

Name: Dart Thrower's Motion

Region: Wrist

Phase: Wrist rehab post injury

Purpose: Functional wrist oblique plane motion

Technique: Wrist extension/radial deviation to flexion/ulnar deviation arc

Dosage: 2 sets 15

References: see PubMed [dart thrower motion wrist rehab]

RECORD: Wall Slide / Scapular Plane Elevation

ID: EX-UE-010

Name: Scapular Plane Wall Slides

Region: Shoulder

Phase: Early–intermediate

Purpose: Elevate with scapular control

Technique: Forearms on wall, slide up in scapular plane

Dosage: 2 sets 10–12

References: see PubMed [scapular plane elevation rehab]

RECORD: Sleeper Stretch (Posterior Capsule)

ID: EX-UE-011

Name: Sleeper Stretch

Region: Shoulder

Phase: Throwing athletes GIRD

Purpose: Posterior capsule mobility IR gain

Technique: Side-lying, gentle IR stretch at shoulder 90° abduction

Dosage: 3 x 30 sec

Precautions: Avoid anterior shoulder pain

References: see PubMed [GIRD sleeper stretch throwers]

RECORD: Wrist Flexor Eccentric (Golfer's Elbow)

ID: EX-UE-012

Name: Eccentric Wrist Flexion

Region: Medial elbow

Phase: Medial epicondylalgia

Purpose: Load common flexor tendon

Technique: Slow eccentric wrist flexion with dumbbell

Dosage: 3 x 15

References: see PubMed [medial epicondylalgia eccentric]

RECORD: Prone Row / Horizontal Pull

ID: EX-UE-013

Name: Prone Horizontal Row

Region: Shoulder/scapula

Phase: Intermediate

Purpose: Scapular retraction strength

Technique: Prone, row band/dumbbell elbow extended shoulder

Dosage: 3 x 10–12

References: see PubMed [scapular strengthening row]

RECORD: Push-up Plus (Serratus Anterior)

ID: EX-UE-014

Name: Push-up Plus

Region: Scapulothoracic

Phase: Intermediate

Purpose: Serratus anterior activation

Technique: Push-up position, protract scapulae at top

Dosage: 2–3 sets 10–15

References: see PubMed [push-up plus serratus]

RECORD: Finger Tendon Glides

ID: EX-UE-015

Name: Tendon Glides (Flexor)

Region: Hand

Phase: Post flexor injury/CTS

Purpose: Maintain tendon glide

Technique: Hook, full fist, straight fist, flat hand sequential

Dosage: 10 reps each position 3–5x daily

References: see PubMed [flexor tendon glides Kleinert]

RECORD: Pronation/Supination with Band

ID: EX-UE-016

Name: Resisted Forearm Pronation/Supination

Region: Forearm/elbow

Phase: Intermediate

Purpose: Forearm rotator strength

Technique: Band resistance rotation elbow 90°

Dosage: 3 x 12 each direction

References: see PubMed [forearm pronation supination rehab]

13. Return to Sport

RECORD: Return to Tennis — Lateral Epicondylalgia

ID: RTS-UE-001

Sport: Tennis

Condition: Lateral epicondylalgia

Criteria: Pain-free grip >85%, Cozen negative, can simulate forehand/backhand/serve without pain

Progression: Groundstrokes → volleys → serve progression over 2–4 weeks

Timeline: 6–12 weeks minimum post symptom resolution

References: see PubMed [tennis elbow return to sport]

RECORD: Return to Baseball/Throwing — UCL / Elbow

ID: RTS-UE-002

Sport: Baseball/throwing

Condition: UCL sprain or post-reconstruction

Criteria: Full elbow ROM, valgus stress stable, interval throwing program stages complete

Progression: ASMI interval throwing: 45 ft → long toss → mound

Timeline: UCL reconstruction 12–18 months; sprain variable 6–12 weeks

References: ASMI interval throwing program; see PubMed [UCL return to throw]

RECORD: Return to Golf — Medial Epicondylalgia / Wrist

ID: RTS-UE-003

Sport: Golf

Condition: Medial epicondylalgia, wrist tendinopathy

Criteria: Pain-free swing simulation, grip strength symmetric

Progression: Putting → chipping → full swing gradual volume

Timeline: 4–8 weeks post pain-free strengthening

References: see PubMed [golf elbow return to play]

RECORD: Return to Climbing / Grip Sports — Pulley / Finger

ID: RTS-UE-004

Sport: Rock climbing / grip sports

Condition: Finger pulley strain, TFCC, elbow tendinopathy

Criteria: Pain-free hang test sub body weight, grip >90%, no morning stiffness

Progression: Easy routes → progressive load → full grade over 4–8 weeks

Timeline: Pulley grade I–II 4–6 weeks; TFCC 8–12+ weeks

References: see PubMed [climbing finger injury return to sport]

14. Imaging

RECORD: X-ray (Radiography) — Upper Extremity

ID: IMG-UE-001

Modality: X-ray

Indications: Fracture, dislocation, arthritis, calcific tendinopathy, malalignment

Typical Findings: Fracture lines, joint space narrowing, osteophytes, effusion fat pad sign elbow

Sensitivity_Specificity: Fracture high sensitivity if views adequate; scaphoid may need dedicated views

Limitations: Soft tissue, early stress injury, ligament/tendon

References: see PubMed [upper extremity radiography indications]

RECORD: MRI — Upper Extremity

ID: IMG-UE-002

Modality: MRI

Indications: Soft tissue — rotator cuff, labrum, UCL, TFCC, SL ligament, tumor, infection

Typical Findings: Tendon tear signal, ligament discontinuity, bone edema, nerve compression

Sensitivity_Specificity: RC full tear sens ~90%; SL ligament improved with arthrogram

Limitations: Cost, motion artifact, false positive degenerative findings

References: see PubMed [MRI shoulder elbow wrist indications]

RECORD: Ultrasound — Upper Extremity

ID: IMG-UE-003

Modality: Ultrasound

Indications: Tendinopathy, rotator cuff, nerve entrapment, dynamic tendon subluxation, guided injection

Typical Findings: Hypoechoic tendon, partial tear, nerve swelling CTS

Sensitivity_Specificity: Operator dependent; cuff tear comparable MRI experienced hands

References: see PubMed [musculoskeletal ultrasound upper extremity]

RECORD: CT — Upper Extremity

ID: IMG-UE-004

Modality: CT

Indications: Complex fracture, glenoid/scaphoid assessment, preoperative planning, hook hamate

Typical Findings: Fracture comminution, articular step-off, bone healing

Limitations: Radiation, limited soft tissue

References: see PubMed [CT upper extremity trauma]

15. Evidence & Guidelines

RECORD: APTA Clinical Practice Guidelines — Upper Extremity

ID: EVD-UE-001

Source: APTA / Academy of Orthopaedic Physical Therapy

Topic: Shoulder pain, manual therapy, exercise

Evidence Level: Level A–B depending on recommendation

Summary: Supports exercise and manual therapy for shoulder conditions; individualized approach

Reference Note: see PubMed [APTA shoulder clinical practice guideline]

RECORD: AAOS Clinical Practice Guideline — Carpal Tunnel Syndrome

ID: EVD-UE-002

Source: American Academy of Orthopaedic Surgeons

Topic: CTS diagnosis and treatment

Evidence Level: Level A

Summary: Recommends EMG for confirmation; local steroid injection and surgical release effective

Reference Note: AAOS CTS CPG 2016; see PubMed [AAOS carpal tunnel guideline]

RECORD: Cochrane Review — Tennis Elbow (Lateral Epicondylalgia)

ID: EVD-UE-003

Source: Cochrane Library

Topic: Interventions lateral epicondylitis

Evidence Level: Level A meta-analysis

Summary: Eccentric exercise, PT superior long-term vs corticosteroid alone; CSI short-term relief only

Reference Note: PMID 23813461 (conceptual); see PubMed [Cochrane tennis elbow]

RECORD: ASMI — Interval Throwing Program

ID: EVD-UE-004

Source: American Sports Medicine Institute

Topic: Return to throwing after injury

Evidence Level: Level C expert consensus

Summary: Structured distance/pitch count progression standard of care throwers

Reference Note: ASMI interval throwing program document

RECORD: Clinical Practice Guideline — Rotator Cuff

ID: EVD-UE-005

Source: AAOS / AOSSM literature

Topic: Rotator cuff tears

Evidence Level: Level B

Summary: Conservative PT reasonable partial tears; surgery for full thickness active patients failing rehab

Reference Note: see PubMed [rotator cuff tear management guideline]